Integrated Passive Seismic Constraints on Sedimentary and Crustal Structure across the Southern Appalachian and Atlantic Coastal Plain: Part 1

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Abstract

Accurate characterization of shear-wave velocity (V_s) structure beneath sedimentary basins is essential for seismic hazard assessment, yet regional constraints in the southeastern United States are limited by the non-uniqueness of any single passive observable, and no previous regional study has jointly exploited microtremor horizontal-to-vertical spectral ratios (HVSR), Rayleigh-wave ellipticity, and ambient-noise surface-wave dispersion within a unified theoretical framework. We synthesize these observables within the established diffuse-field framework and apply them to continuous broadband recordings from the SESAME array spanning the southern Appalachians and the Atlantic Coastal Plain (ACP). Under diffuse-field conditions, HVSR can be written as a ratio of Green's function components, and under fundamental-mode Rayleigh-wave dominance this expression reduces to Rayleigh-wave ellipticity, providing a direct physical link between the two observables.

We confirm this prediction observationally by showing spectral continuity between ellipticity (3–25 s; 0.04–0.33 Hz) and microtremor HVSR (0.2–25 Hz) in the overlap band near 0.3 Hz, and together HVSR and ellipticity provide a continuous site-response characterization from approximately 0.04 to 25 Hz. We apply an amplitude-preserving cross-correlation workflow that retains the inter-component amplitude ratios needed for ellipticity extraction, and we introduce a reduced chi-square quality control metric for HVSR processing that identifies and removes anomalous hourly records. HVSR fundamental frequencies span more than an order of magnitude across the array and closely parallel sediment-thickness contours on the eastern transects, confirming that the sediment–basement impedance contrast is the primary control on site resonance. On the western array, however, anomalously high resonance frequencies despite thick sediments reveal an east–west structural divergence that we attribute to the presence of stiffer material within the sedimentary column acting as an effective seismic basement above the true crystalline basement. New high-resolution phase and group velocity maps resolve the sharp velocity gradient across the Fall Line at finer spatial detail than continental-scale tomographic models, with short-period group velocities showing a north–south contrast exceeding 1 km/s, and Rayleigh-wave ellipticity maps at 4 s and 17 s capture the period-dependent transition from shallow sediment-controlled to deeper crust-controlled structure. A preliminary forward-modelling test demonstrates that a single five-layer velocity model simultaneously satisfies all four observables at a representative Coastal Plain station, validating the theoretical framework and establishing the observational foundation for the nonlinear Bayesian joint inversion developed in Part 2 of this study.

Plain Language Summary

Earthquake shaking can be strongly amplified in regions where cities are built on thick, soft sediments. Predicting this amplification requires knowing how seismic-wave speeds change with depth in the sediments and

the rocks below. We analyzed continuous recordings of natural ground vibrations (from ocean waves, weather, and human activity) from the SESAME seismometer array across the southern Appalachian Mountains and the Atlantic Coastal Plain.

From these recordings, we extracted three different types of measurements that each reveal different aspects of the underground structure: the natural resonance frequency of the sediment layer at each station, the shape of seismic wave motion at the surface, and the speed at which seismic waves travel between stations. We showed mathematically and observationally that these measurements are connected through the same physical framework and that they provide complementary information about the subsurface. Our results reveal that sediment thickness is the main factor controlling ground resonance along the eastern part of our study area, but toward the southwestern part of the study area the pattern changes, suggesting that a stiffer layer within the sediments acts as a secondary boundary that affects ground shaking. We also produced detailed maps of seismic wave speed that capture sharp transitions between sediment and rock that previous large-scale studies could not resolve. By demonstrating that one velocity model can explain all of these measurements simultaneously, we establish the foundation for a companion study (Part 2) that will estimate shear-wave velocity structure with uncertainty, improving hazard assessments in the rapidly growing southeastern United States.

1 Introduction

Accurate characterization of shear-wave velocity (V_S) structure beneath sedimentary basins is essential for seismic hazard assessment, as the mechanical properties of the shallow subsurface exert primary control over the amplitude, frequency content, and duration of earthquake ground motions at the surface. Strong impedance contrasts between unconsolidated sediments and the underlying crystalline basement produce resonance within the sedimentary column, trapping and amplifying seismic energy at frequencies determined by both the total sediment thickness and the vertical velocity gradient (Borcherdt, 1970, 1994; Bard and Bouchon, 1985). This effect is particularly consequential in densely populated urban regions, many of which overlie thick sedimentary basins whose internal geometry, lateral heterogeneity, and depth-dependent material properties govern the intensity and spatial variability of ground shaking during earthquakes (Olsen, 2000; Lai et al., 2020). Deep basin structures efficiently amplify long-period ground motions with periods greater than 1 second, which are especially damaging to tall buildings, long-span bridges, and other critical infrastructure with natural frequencies in this range (Kawase, 1996; Joyner, 2000; Moschetti et al., 2017; Moczo et al., 2018). Amplification results from seismic energy becoming trapped within low-velocity sedimentary fill, where repeated reflection, refraction, and diffraction at subsurface velocity discontinuities produce constructive interference that sustains and intensifies ground shaking beyond levels expected for uniform bedrock (Graves et al., 1998; Frankel et al., 2009; Petersen et al., 2024). Robust seismic hazard evaluation in these environments requires well-constrained, three-dimensional V_S models that resolve the velocity structure of the entire sedimentary column from surface to basement. The necessity of incorporating basin effects into hazard calculations is highlighted by the 2023 update to the U.S. National Seismic Hazard Model (Petersen et al., 2024), which for the first time included refined V_S basin characterizations for several major urban areas including Seattle, the San Francisco Bay area, greater Los Angeles, and the Salt Lake City region to improve predictions of long-period ground shaking where shallow geology departs from reference rock conditions (Boore et al., 2014; Campbell and Bozorgnia, 2014).

In the southeastern United States, the southern Appalachian Mountains and the adjacent Atlantic Coastal Plain (ACP) together present a compelling target for multi-scale shear-wave velocity characterization (Figure 1a). The Appalachians expose crystalline basement with complex internal structure inherited from multiple orogenic events, while the ACP and related Mississippi Embayment are underlain by a broad wedge of Cretaceous and younger sediments whose strata cover large areas of the central and eastern United States (Pratt et al., 2025). Sediment thickness varies from zero at the Fall Line, the inland limit of the Coastal Plain, to more than 3 km along the Georgia and South Carolina coast and exceeds 10 km beneath the Florida Platform and Gulf margin (Figure 1a). The northern ACP is commonly dominated by clastic facies sourced from Appalachian erosion, whereas carbonate-dominated sequences become more prominent southward toward the Florida Platform (Renken, 1996). These flat-lying strata are considerably slower and less dense than the underlying Paleozoic basement, creating sharp rheological and seismic impedance contrasts

(Lawrence and Hoffman, 1993). Across much of the wedge, shear-wave velocity increases with depth owing to compaction and lithification: near-surface materials may exhibit V_S on the order of a few hundred m/s, increasing to approximately 600 to 1000 m/s in deeper sediments, followed by a sharp increase to multi-km/s values in the crystalline basement (Chapman et al., 2003; Pratt et al., 2017). Strong velocity gradients and impedance contrasts produce resonance whose fundamental frequency depends on sediment thickness and average V_S structure, while higher-frequency features can reflect internal layering within the sedimentary column (Pratt et al., 2017; Cunningham and Lekic, 2020).

The ACP sedimentary wedge overlies a tectonically complex basement assembled through multiple Paleozoic orogenies and later modified by Mesozoic rifting. The Taconic (480–440 Ma), Acadian (420–380 Ma), Neoacadian (380–340 Ma), and Alleghanian (330–260 Ma) events collectively built the southern Appalachian orogen through accretion and collision, producing a mosaic of terranes and suture zones that continue beneath the ACP (Hatcher et al., 2007; Hibbard et al., 2010). The boundary between Laurentian-Appalachian crust and the exotic Gondwanan Suwannee terrane is marked by the Suwannee suture, which roughly coincides with the Fall Line and the Higgins–Zietz magnetic boundary. South of this boundary, crustal thinning and syn-rift basin development during the Mesozoic (e.g., the South Georgia Rift) have left the Coastal Plain with a thinner crust and deep buried basins. Seismic reflection profiling has imaged major detachment systems and deep crustal structures associated with Alleghanian convergence (Nelson et al., 1985; Cook and Vasudevan, 2006), while later passive-source studies further refined the inferred geometry of sutures and overthrusting. This tectonic inheritance is relevant to both hazard and structure because lateral variations in basement composition and crustal architecture modulate sediment accumulation patterns and seismic wave propagation.

A key consequence of this geological structure is the amplification of long-period (>1 s) ground motions within the soft sedimentary column, an effect not adequately captured by the widely used metric V_{S30}, which is the time-averaged shear-wave velocity in the upper 30 meters (Borcherdt, 1994). The assumption that the uppermost 30 meters reliably represent deeper subsurface conditions fails when sediment thickness extends to kilometer-scale depths (Chapman and Guo, 2021): in the ACP, strong impedance contrasts at the sediment-basement boundary trap seismic energy through internal reflections and generate resonance that depends on total sediment thickness rather than near-surface properties alone (Pratt et al., 2017; Pratt and Schleicher, 2021). The practical significance of this deep amplification is well documented. During the 2011 M_w 5.8 Mineral, Virginia, earthquake, ACP sediments amplified moderate ground motions to damaging levels in Washington, D.C., more than 130 km from the epicenter (Hough, 2012; Wells et al., 2015; Pratt et al., 2017), and strong shaking was again widely reported during the 2024 M_w 4.8 Tewksbury, New Jersey, earthquake, for which three-dimensional ground-motion simulations show that nearby sedimentary basins, including Coastal Plain strata, strongly modulate the long-period shaking (Boyd et al., 2026). ACP sediments also likely contributed to the destructive effects of the 1886 M_w 7 Charleston, South Carolina, earthquake (Chapman et al., 2006), and the Charleston seismic zone remains active, with paleoliquefaction evidence indicating six major earthquakes in the past 6,000 years (Talwani and Schaeffer, 2001; Chapman et al., 2016). Further inland, the Eastern Tennessee Seismic Zone demonstrates that intraplate seismicity can concentrate in basement structures below the Appalachian décollement (Levandowski et al., 2018). These ongoing hazards underscore the need for improved constraints on velocity structure from the shallow subsurface through the crust, spanning both the Appalachian crystalline terrane and the adjacent Coastal Plain.

Decades of active-source and passive seismic experiments have progressively refined aspects of regional crustal shear-wave velocity structure, yet important gaps remain in linking shallow sediments to deeper crust within a single consistent framework. The SESAME deployment (Figure 1b) provided new passive constraints on crustal thickness and intra-crustal boundaries, including evidence for a preserved crustal root beneath the Blue Ridge and lateral changes in crustal properties (Parker et al., 2013). Active-source profiles and receiver function studies indicate that crustal thickness decreases from approximately 45 to 56 km in the Inner Piedmont and Blue Ridge to approximately 29 to 38 km beneath the Coastal Plain near the Atlantic coast, consistent with extension during the Mesozoic opening of the Atlantic (Marzen et al., 2019). The Moho beneath the ACP is relatively flat, with only modest relief of a few kilometres, whereas under the Appalachian foothills it deepens into a root beneath the Blue Ridge Mountains. This transition occurs across the ancient suture zone, and the juxtaposition of thick Appalachian crust against thinned rifted-margin crust may help localize intraplate seismicity in the region. Prior efforts using USArray data provided broad-scale

velocity models for the southeastern United States crust and upper mantle, but lacked the resolution to delineate fine-scale sediment stratigraphy within the Coastal Plain or resolve intra-crustal velocity variations across the Appalachian-to-Coastal Plain transition (Marzen et al., 2019; Hopper et al., 2017). These existing studies often emphasize either deep crustal structure beneath the Appalachians or localized near-surface site conditions on the Coastal Plain, leaving the intermediate-depth sedimentary column (hundreds of meters to several kilometers) and the lateral transition in crustal properties across the Fall Line comparatively underconstrained despite their importance for both hazard assessment and basin architecture.

Joint inversion of passive seismic observables offers an efficient pathway to bridge this gap. The horizontal-to-vertical spectral ratio (HVSR) method, introduced by Nogoshi and Igarashi (1970, 1971) and popularized by Nakamura (1989), estimates the fundamental site-resonance frequency f_0 from single-station ambient vibration recordings and is widely used for site characterization (Bonnefoy-Claudet et al., 2006; Nakamura, 2019). Although HVSR measurements are rapid and provide robust estimates of f_0 where impedance contrasts are strong, their interpretation can be ambiguous because different combinations of layer thickness and velocity may produce similar resonance frequencies, and the method alone does not constrain absolute velocities or structures deeper than the dominant impedance interface. Surface-wave dispersion from ambient noise cross-correlation provides complementary constraints on how velocities change with depth (Bensen et al., 2007). Studies using USArray Transportable Array data have consistently found very low shear velocities in the upper crust beneath the Coastal Plain, attributable to the thick column of unconsolidated sediments (Yang et al., 2021; Ekström, 2017; Shen and Ritzwoller, 2016). However, dispersion data alone exhibit trade-offs between thickness, velocity, and gradient structure that can leave sharp boundaries and thin layers poorly resolved. Rayleigh-wave ellipticity, the frequency-dependent ratio of horizontal-to-vertical Rayleigh-wave particle motion at the surface, provides a third complementary constraint that is particularly sensitive to shallow V_S structure and impedance contrasts (Fäh et al., 2001; Lin et al., 2012, 2014). Mi and Xia (2023) demonstrated that ellipticity inversion succeeds even for models with large velocity contrasts between unconsolidated sediments and high-velocity bedrock, underscoring its value as a supplement to dispersion for resolving strong shallow contrasts. Combining these observables helps to constrain sediment thickness, velocity gradients, and impedance contrasts far more tightly than any single dataset can achieve alone (Scherbaum et al., 2003; Wathelet et al., 2020; Pratt et al., 2025). Earlier joint inversions of HVSR and dispersion curves have estimated shallow V_S structure in sedimentary settings (e.g., Arai and Tokimatsu, 2004; Parolai et al., 2004; Neukirch et al., 2021; Rotimi et al., 2025; Zhang et al., 2025), but most have resolved only the upper $\sim$5 km, and no previous regional study in the southeastern United States has jointly exploited ambient-noise HVSR, Rayleigh-wave ellipticity, and noise-derived surface-wave dispersion.

In this work, we use HVSR, surface-wave dispersion (phase and group velocity), and Rayleigh-wave ellipticity from ambient noise to characterize the sedimentary and crustal structure from the southern Appalachians across the Fall Line transition into the deep Coastal Plain. Our objectives are to (1) synthesize a unified theoretical framework that connects microtremor HVSR, Rayleigh-wave ellipticity, and surface-wave dispersion as complementary constraints on the same underlying velocity structure, (2) develop and apply robust preprocessing, processing, and quality control procedures that extract reliable measurements of all three observables from the SESAME broadband array, and (3) demonstrate through forward modelling that these observables are mutually consistent and collectively provide the observational foundation needed for basin-scale shear-wave velocity imaging from shallow sediments through the deeper crust. Hence, this study is presented in two parts. Part 1, presented here, synthesizes the theoretical basis unifying the observables under the diffuse-field assumption, describes the processing and quality control procedures we developed and applied, and presents regionalized phase- and group-velocity dispersion maps alongside continuous station-level ellipticity and HVSR characterizations, revealing fundamental lithological transitions within the region's sedimentary architecture. Part 2 will develop and implement a nonlinear hierarchical Bayesian joint inversion scheme that combines all observables to determine the shear-wave velocity structure from the near surface to the upper mantle with full uncertainty quantification. We separate the inversion into its own article because no single one-dimensional velocity model can satisfy observables that vary from station to station and from transect to transect, and the ensemble Markov chain Monte Carlo framework required to explore station-wise model populations demands a development and validation effort of its own.

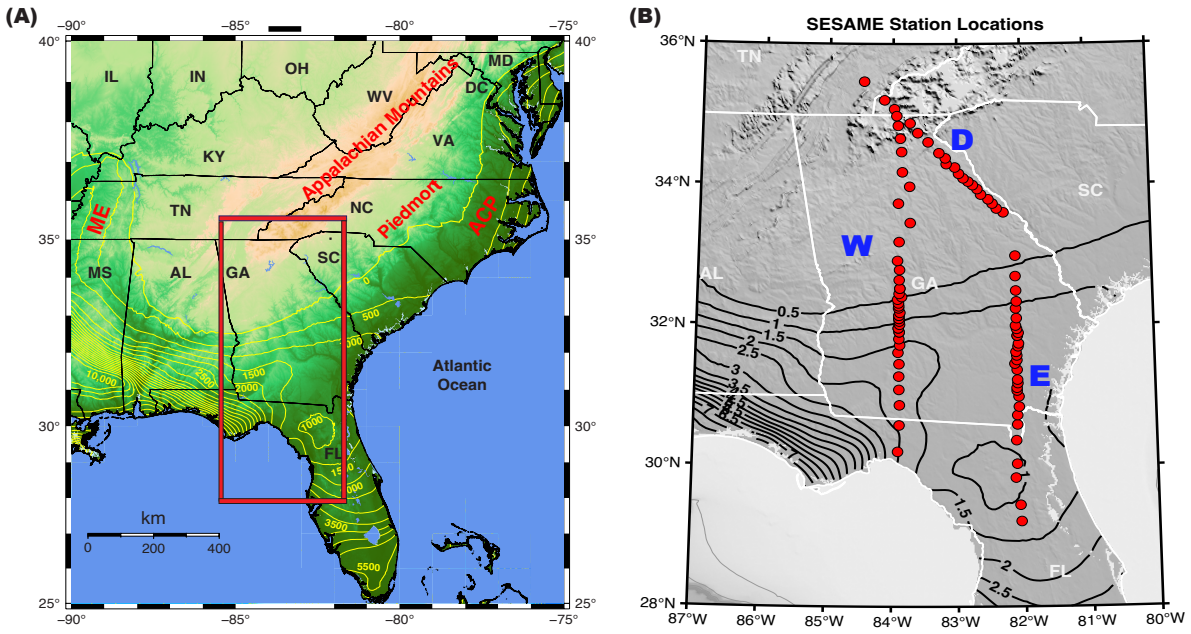

Figure 1: (A) Topographic map of the southeastern United States showing the major physiographic provinces: Appalachian Mountains, Piedmont, Atlantic Coastal Plain (ACP), and Mississippi Embayment (ME). Yellow contours indicate sediment thickness in meters. The red box outlines the SESAME study area. (B) SESAME station locations (red circles) within the red box with sediment thickness contours in kilometers. The three principal transects are labeled: D (diagonal), W (western), and E (eastern).

2 Data

We analyze continuous broadband recordings from the 88-station SESAME (Southeastern Suture of the Appalachian Margin Experiment) array, a temporary deployment that operated from 2010 to 2014 across Georgia, South Carolina, and portions of adjacent states including North Carolina, Tennessee, Alabama, and Florida (Fischer et al., 2010). The array spans latitudes ∼28°N to 36°N and longitudes ∼80°W to 87°W, and is organized along three principal transects (Figure 1b): a diagonal transect (D) extending from the northern Piedmont southeastward across the Fall Line into South Carolina, a western transect (W) running north–south through central Georgia from the Piedmont into the deep Coastal Plain, and an eastern transect (E) running north–south along the Georgia–South Carolina coastal margin. This geometry samples the full depth range of the ACP sedimentary wedge, from zero thickness on the Piedmont crystalline outcrop to more than 7 km in the southwestern portion of the study area, and its station spacing (approximately 9–50 km) provides higher lateral resolution than the USArray Transportable Array (∼70 km) across the Appalachian-to-Coastal-Plain transition.

Stations were equipped with broadband seismometers (Nanometrics Trillium 120PA and Streckeisen STS-2) recording continuously at 100 samples per second. Installations included both vault-mounted concrete piers and shallow posthole emplacements; prior testing demonstrated that these configurations do not introduce systematic differences relevant to the spectral ratios and cross-correlation measurements used here, and continuous waveform data were retrieved from the IRIS Data Management Center for the full deployment interval.

3 Methods

3.1 Theoretical basis for unifying microtremor HVSR and Rayleigh-wave ellipticity under the Diffuse Field Assumption

This section synthesizes established results from the diffuse-field literature into the framework we apply throughout the study. We first define HVSR and review its physical interpretations (Section 3.1.1) and express it through component power and directional energy density (Section 3.1.2). Following Sánchez-Sesma et al. (2011), we then cast HVSR as a ratio of Green's-function components under the diffuse-field assumption (Section 3.1.3) and define Rayleigh-wave ellipticity as a mode-specific observable (Section 3.1.4). Using the modal decomposition of Sánchez-Sesma et al. (2011) and García-Jerez et al. (2016), we show that HVSR reduces to fundamental-mode Rayleigh-wave ellipticity under conditions commonly met near the principal site resonance (Section 3.1.5), and we close by evaluating whether the ACP wavefield satisfies the diffuse-field conditions required for this convergence (Section 3.1.6).

3.1.1 HVSR: definition and physical interpretations

The horizontal-to-vertical spectral ratio (HVSR) is a single-station observable computed from three-component ground-motion recordings, and in the standard ambient-noise implementation, the ratio of horizontal to vertical Fourier spectral amplitudes is averaged over multiple time windows to yield a stable, frequency-dependent H/V curve. Following Nakamura (1989), HVSR has become widely adopted for site characterization because it provides a robust estimate of the fundamental site-resonance frequency f_0 with minimal instrumentation, and the processing guidelines are codified in the SESAME recommendations (Bard and SESAME participants, 2005).

Two principal frameworks have been advanced for interpreting HVSR. The first links the H/V curve to Rayleigh-wave ellipticity, the frequency-dependent horizontal-to-vertical particle-motion ratio governed by shallow elastic structure (e.g., Lermo and Chávez-García, 1994; Malischewsky and Scherbaum, 2004), and the second treats the H/V peak as an indicator of S-wave resonance in horizontally layered media (e.g., Nakamura, 1989, 2000). Both interpretations are complicated by the fact that the ambient wavefield generally comprises unknown proportions of Rayleigh waves, Love waves, and body waves, so the observed HVSR reflects not only subsurface structure but also wavefield composition (Fäh et al., 2001; Bonnefoy-Claudet et al., 2006; Lunedei and Malischewsky, 2015).

A more complete treatment is provided by the diffuse-field assumption (DFA), under which the ambient wavefield is modelled as a stationary superposition of multiply scattered waves that is sufficiently randomized (ideally equipartitioned) across propagation directions and wave types. In this regime HVSR becomes expressible through the imaginary part of the elastodynamic Green's tensor, yielding a full-wave structural interpretation that does not require Rayleigh-wave dominance (Sánchez-Sesma et al., 2011; García-Jerez et al., 2016; Piña-Flores et al., 2017). This framework motivates the formulation we summarize below.

3.1.2 HVSR expressed through component power and its connection to directional energy density

For a receiver at position $\mathbf{x}$ and angular frequency ω ($f = \omega/2\pi$), the window-averaged HVSR is

$$\mathrm{HVSR}(\mathbf{x},\omega) = \left[\frac{\langle|u_1(\mathbf{x},\omega)|^2\rangle + \langle|u_2(\mathbf{x},\omega)|^2\rangle}{\langle|u_3(\mathbf{x},\omega)|^2\rangle}\right]^{1/2} = \left[\frac{E_1(\mathbf{x},\omega) + E_2(\mathbf{x},\omega)}{E_3(\mathbf{x},\omega)}\right]^{1/2}, \tag{1}$$

where u_1 and u_2 are horizontal displacement spectra in two orthogonal directions, u_3 is the vertical component, $\langle\cdot\rangle$ denotes multi-window averaging, and E_i is the directional energy density (DED) for component i. Under the diffuse-field framework the DED is (Sánchez-Sesma et al., 2011; Piña-Flores et al., 2017)

$$E_i(\mathbf{x},\omega) = \rho(\mathbf{x})\,\omega^2\,\langle|u_i(\mathbf{x},\omega)|^2\rangle, \tag{2}$$

where ρ is local mass density. Because ρ and ω^2 are common to all three components at a co-located receiver, they cancel in the spectral ratio; the DED formulation is therefore algebraically equivalent to the

displacement-power ratio but connects the observable to the energy-density language of diffuse-field theory developed in Section 3.1.3.

For time–frequency analysis, we compute a windowed HVSR:

$$\mathrm{HVSR}(t,f) = \left[\frac{E_1(t,f) + E_2(t,f)}{E_3(t,f)}\right]^{1/2} \tag{3}$$

from which we identify f_0 following Bard and SESAME participants (2005).

3.1.3 Diffuse-field theory: HVSR as a Green's-function ratio

Under the DFA, displacement cross-correlations are proportional to the imaginary part of the elastodynamic Green's tensor (Weaver, 1982; Sánchez-Sesma et al., 2011):

$$\left\langle u_i(\mathbf{x}_A,\omega)\, u_j^*(\mathbf{x}_B,\omega)\right\rangle \;\propto\; \Im\big[G_{ij}(\mathbf{x}_A,\mathbf{x}_B;\omega)\big], \tag{4}$$

where G_{ij} is the displacement in direction i at $\mathbf{x}_A$ due to a unit harmonic point load in direction j at $\mathbf{x}_B$. The azimuthal averaging implicit in Eq. (4) is a consequence of equipartition rather than an additional processing step. Setting $\mathbf{x}_A = \mathbf{x}_B = \mathbf{x}$ and $i = j$, the DEDs become proportional to $\Im(G_{ii})$, giving the diffuse-field HVSR (Sánchez-Sesma et al., 2011):

$$\mathrm{HVSR}_{\mathrm{DFA}}(\mathbf{x},\omega) = \left[\frac{\Im\big(G_{11}(\mathbf{x},\mathbf{x};\omega)\big) + \Im\big(G_{22}(\mathbf{x},\mathbf{x};\omega)\big)}{\Im\big(G_{33}(\mathbf{x},\mathbf{x};\omega)\big)}\right]^{1/2}. \tag{5}$$

Equation (5) casts HVSR as a structural response function determined by local elastic properties through the Green's tensor. Under equipartition, Rayleigh waves, Love waves, and body waves all enter through $\Im(G_{ii})$, so no assumption of Rayleigh-wave dominance is required.

When the ambient wavefield is equipartitioned, the autocorrelation spectrum at a single station maps directly onto the imaginary part of the local Green's tensor, so the conventional HVSR of Eq. (1) and the Green's-function ratio of Eq. (5) become formally equivalent (Sánchez-Sesma et al., 2011), an equivalence confirmed empirically for ambient-noise recordings (Spica et al., 2018; Perton et al., 2020) and for earthquake coda under sufficiently diffuse conditions (Salinas et al., 2014). Importantly, this equivalence does not require that the noise field be perfectly uniform in source azimuth at all times; it requires that the wavefield be sufficiently randomized over the averaging interval so that stable component energy ratios reflect the site's structural response. The degree to which the DFA holds in practice depends on the geological setting, and in environments with limited scattering heterogeneity, for example the East Antarctica ice sheet studied by Sivaram et al. (2025), the wavefield may not be fully equipartitioned and the equivalence between HVSR and $\Im(G_{ii})$ must be treated with caution. Departures from diffuseness or equipartition can therefore bias observed H/V curves, and we evaluate in Section 3.1.6 the conditions under which the DFA, or a weaker practically sufficient form of it, is satisfied in the Atlantic Coastal Plain.

3.1.4 Rayleigh-wave ellipticity (H/V)

Rayleigh-wave ellipticity characterises the polarisation of Rayleigh-wave particle motion and is defined as the horizontal-to-vertical amplitude ratio of the Rayleigh-wave particle motion at the free surface. For mode n it is defined as

$$\varepsilon_R^{(n)}(\omega) \equiv \frac{|r_1^{(n)}(0,\omega)|}{|r_2^{(n)}(0,\omega)|}, \tag{6}$$

where $r_1^{(n)}$ and $r_2^{(n)}$ are the horizontal and vertical Rayleigh eigenfunctions at $z = 0$ for mode n. We adopt the H/V convention in this article, while the inverse quantity $ZH \equiv 1/\varepsilon_R$ is used by Tanimoto et al. (2013).

The shape of $\varepsilon_R^{(n)}(\omega)$ is controlled primarily by shallow V_s layering: a strong impedance contrast between a low-velocity surface layer and stiffer basement tends to produce a pronounced ellipticity peak near the fundamental resonance frequency, while subsequent minima reflect multi-layer structure (Malischewsky

and Scherbaum, 2004). This sensitivity makes ellipticity a powerful complement to dispersion data for constraining shallow velocity structure (Fäh et al., 2003; Arai and Tokimatsu, 2004; Hobiger et al., 2013).

In our work we focus on the fundamental mode,

$$\varepsilon_{R,0}(\omega) \equiv \varepsilon_R^{(n=0)}(\omega) = \frac{|r_{1,0}(0,\omega)|}{|r_{2,0}(0,\omega)|}, \tag{7}$$

which dominates the ambient-wavefield response near the principal site resonance in most sedimentary settings.

3.1.5 Unifying HVSR and Rayleigh-wave ellipticity through modal decomposition

Equations (5) and (6) describe two distinct observables: HVSR encodes the full-wave partition of horizontal and vertical energy under diffuse-field conditions, while ellipticity isolates mode-specific Rayleigh-wave polarization. To make their connection explicit, we follow the normal-mode formulation of Sánchez-Sesma et al. (2011) and García-Jerez et al. (2016) and decompose $\Im(G_{ii})$ into modal contributions.

Modal decomposition of $\Im(G_{ii})$: In a laterally homogeneous, layered elastic half-space, $\Im(G_{ii})$ at a co-located source–receiver separates into discrete surface-wave modes and a body-wave continuum (Sánchez-Sesma et al., 2011; Piña-Flores et al., 2017). Love waves carry no vertical displacement, so only Rayleigh modes and body waves contribute to the vertical component:

$$\Im\big[G_{33}(\mathbf{x},\mathbf{x};\omega)\big] = \sum_{n=0}^{N_R} \frac{r_{2,n}^2(0;\omega)}{8\,c_n\,U_n\,I_n} + \mathcal{B}_{33}(\omega), \tag{8}$$

where c_n, U_n, and I_n are the phase velocity, group velocity, and energy integral for Rayleigh mode n, and $\mathcal{B}_{33}$ collects body-wave contributions. For the horizontal components, the azimuthal isotropy of the diffuse field ensures $\Im(G_{11}) = \Im(G_{22})$, and both Rayleigh and Love modes enter:

$$\Im\big[G_{11}+G_{22}\big] = \sum_{n=0}^{N_R} \frac{r_{1,n}^2(0;\omega)}{8\,c_n\,U_n\,I_n} + \sum_{m=0}^{N_L} \frac{l_{1,m}^2(0;\omega)}{8\,c_m^L\,U_m^L\,I_m^L} + \mathcal{B}_H(\omega), \tag{9}$$

where $l_{1,m}$ is the Love eigenfunction at the surface for mode m, superscript L marks Love-mode parameters, and $\mathcal{B}_H$ is the horizontal body-wave term. The azimuthal weighting factors implicit in Eqs. (8) and (9) follow the normalization of García-Jerez et al. (2016).

Full-wave DFA expression: Substituting Eqs (8)–(9) into Eq. (5) yields:

$$\mathrm{HVSR}^2_{\mathrm{DFA}}(\omega) = \frac{\displaystyle\sum_{n=0}^{N_R} \frac{r_{1,n}^2(0)}{8\,c_n\,U_n\,I_n} + \sum_{m=0}^{N_L} \frac{l_{1,m}^2(0)}{8\,c_m^L\,U_m^L\,I_m^L} + \mathcal{B}_H}{\displaystyle\sum_{n=0}^{N_R} \frac{r_{2,n}^2(0)}{8\,c_n\,U_n\,I_n} + \mathcal{B}_{33}}. \tag{10}$$

This expression shows that the diffuse-field HVSR differs in general from any single Rayleigh-mode ellipticity because of Love-wave, higher-mode Rayleigh, and body-wave contributions.

Fundamental-mode Rayleigh-wave approximation: Equation (10) reduces to fundamental-mode Rayleigh ellipticity under two conditions commonly met near the principal site resonance: (i) the $n = 0$ Rayleigh term carries the dominant surface-wave energy so that higher modes are negligible, and (ii) Love-wave and body-wave contributions are small relative to the $n = 0$ Rayleigh term. Under these conditions,

$$\mathrm{HVSR}^2_{\mathrm{DFA}}(\omega) \;\approx\; \frac{r_{1,0}^2(0;\omega)}{r_{2,0}^2(0;\omega)} = \big[\varepsilon_{R,0}(\omega)\big]^2, \tag{11}$$

yielding:

$$\mathrm{HVSR}_{\mathrm{DFA}}(\omega) \;\approx\; \varepsilon_{R,0}(\omega). \tag{12}$$

Equation (12) states that under diffuse-field conditions and in frequency bands where fundamental-mode Rayleigh waves dominate the ambient wavefield, microtremor HVSR reduces to fundamental-mode Rayleigh-wave ellipticity. This approximation is expected to hold primarily near f_0 in settings where the fundamental Rayleigh mode dominates, but outside that band the full modal expression of Eq. (10) should be assumed.

Equation (12) reconciles the Rayleigh-ellipticity interpretation (Lermo and Chávez-García, 1994; Malischewsky and Scherbaum, 2004) with the S-wave-resonance viewpoint (Nakamura, 2000). Under DFA conditions, HVSR is a structural quantity independent of source characteristics, while $\varepsilon_{R,0}$ offers mode-specific sensitivity to impedance contrasts. Away from the fundamental resonance band, where higher modes, Love waves, or body waves contribute appreciably, the full expression (10) applies and $\mathrm{HVSR}_{\mathrm{DFA}}$ will generally exceed $\varepsilon_{R,0}$. A concrete illustration of why this distinction matters comes from the Los Angeles basin, where Tanimoto et al. (2013) showed that conventional H/V estimates of Rayleigh ellipticity can differ systematically by approximately 20% due to contamination by S and Love waves, which elevates the horizontal amplitudes.

3.1.6 Applicability of the diffuse-field assumption in the Atlantic Coastal Plain

The foregoing theory establishes that HVSR carries structural information when the ambient wavefield satisfies the diffuse-field assumption, and that under fundamental-mode Rayleigh-wave dominance, HVSR converges to Rayleigh-wave ellipticity through Eq. (12). Applying this framework to the Atlantic Coastal Plain requires that we address a key concern, the dominant ambient noise source in this region is the Atlantic Ocean. Koper and Burlacu (2015) mapped the fine structure of double-frequency microseisms across North America and demonstrated that strong, directionally coherent energy arrives from the Atlantic, raising the question of whether the DFA can be validly applied to HVSR measurements in a setting with such a pronounced source directionality.

Resolving this question requires a careful distinction between strict equipartition and the weaker conditions actually needed for HVSR to function as a structural observable. Strict equipartition, as formalized by Snieder (2004), demands that energy be distributed equally among all propagating wave types (Rayleigh, Love, and body waves) and across all propagation directions. In a strongly resonant sedimentary basin such as the ACP, this condition is unlikely to hold in full as the sedimentary column concentrates energy into the fundamental-mode Rayleigh wave near the resonance frequency f_0, violating the assumption of equal modal energy distribution.

The diffuse-field framework of Sánchez-Sesma et al. (2011), however, demonstrates that strict equipartition is not a necessary condition for HVSR to carry structural information. In a layered medium, the imaginary part of the Green's tensor, and therefore the HVSR through Eq. (5), is dominated by surface-wave contributions. Hence, what is required is not that all wave types carry equal energy, but rather that the surface-wave field be sufficiently randomized in azimuth so that the Rayleigh-wave ellipticity can be recovered from the spectral ratio. This is a substantially weaker condition than full equipartition, and we argue that the geological setting of the Atlantic Coastal Plain plausibly satisfies it through mechanisms that randomize the wavefield despite the directional oceanic source. Thick, low-velocity sediments bounded above by the free surface and below by higher-velocity basement favor repeated internal reflections, reverberations, and guided-wave propagation (Guo and Chapman, 2019, 2021), and the extensive multipathing that results progressively reduces sensitivity to the initial source direction as the sediments thicken landward from the Fall Line. Irregularities in the basement surface, lateral velocity changes near the Fall Line, and internal sedimentary heterogeneity further scatter incoming energy into a broader range of propagation azimuths at each site (Graves et al., 1998; Kawase, 1996). While ocean microseisms dominate at long periods, anthropogenic and atmospheric sources contribute at higher frequencies and from a wider range of directions, so stacking over long time spans incorporates these diverse illumination azimuths and broadens the effective source distribution further.

Empirical evidence confirms that these mechanisms are sufficient for reliable HVSR measurement in the ACP. Pratt (2018) directly addressed the concern that directional ocean microseisms might invalidate HVSR in this setting by comparing ambient-noise H/V ratios with spectral ratios computed from earthquake recordings across the coastal plain. The fundamental resonance frequencies obtained by the two independent

methods were in close agreement, demonstrating that the strong impedance contrast dominates the spectral ratio and that f_0 is recovered reliably regardless of source azimuth. Earlier work by Bodin and Horton (1999) and Bodin et al. (2001) had already established that ambient-noise H/V measurements in the ACP display a prominent and consistent peak whose frequency is inversely proportional to sediment thickness, and that average shear-wave velocity could be inferred given independent thickness constraints. Similar correlations between H/V peak frequency and basin depth have been documented in other shallow sedimentary settings (Ibs-von Seht and Wohlenberg, 1999; Parolai et al., 2002; Hinzen et al., 2004; Bonnefoy-Claudet et al., 2009), supporting the generality of this relationship.

These observations find a clear explanation within the modal framework of Section 3.1.5. Near the fundamental resonance, the $n = 0$ Rayleigh mode carries the overwhelming majority of surface-wave energy in the ACP sedimentary column. Under these conditions, although strict equipartition in the sense of Snieder (2004) does not hold because energy is concentrated in the fundamental Rayleigh mode rather than distributed equally across all modes and wave types, the weaker DFA conditions required for Eq. (5) are met because the surface-wave field is sufficiently randomized in azimuth by trapping and scattering within the sedimentary wedge. Furthermore, the dominance of the fundamental Rayleigh mode validates the approximation of Eq. (12), under which HVSR reduces to fundamental-mode Rayleigh-wave ellipticity. The ACP therefore represents a geological setting in which the wavefield conditions not only permit the use of HVSR as a structural observable but actively favor the convergence of HVSR and ellipticity, providing mutually consistent constraints on the shallow velocity structure that we exploit in this work.

3.2 Ambient noise cross-correlation and empirical Green's function extraction

Ambient noise tomography exploits the continuous seismic wavefield generated by ocean–seafloor interactions, atmospheric pressure fluctuations, and anthropogenic activity to image subsurface velocity structure without requiring earthquakes or controlled sources. The theoretical foundation is seismic interferometry: cross-correlating long, simultaneous recordings of ambient ground motion at two stations yields a waveform that approximates the elastic impulse response, the empirical Green's function (EGF), between them (Lobkis and Weaver, 2001; Campillo and Paul, 2003; Wapenaar, 2004; Snieder, 2004; Perton et al., 2009). The recovery is exact when noise sources are uniformly distributed over a surface enclosing both receivers, and holds as a practical approximation when (i) the source distribution is broad enough that stationary-phase contributions along the inter-station axis dominate the integral, (ii) intrinsic attenuation is weak or spectral normalization is effective enough that the source power spectrum does not distort the recovered phase, and (iii) the stacking duration is long enough that the time-averaged correlation converges to the ensemble average assumed by the theory (Snieder, 2004; Snieder et al., 2007; Bensen et al., 2007). The cross-correlation is symmetric about zero lag: the positive-lag (causal) side recovers waves propagating from the second station to the first, and the negative-lag (acausal) side recovers waves in the opposite direction. Not all noise sources contribute equally; the dominant contributions arise from stationary-phase regions where sources lie along the extension of the great-circle path connecting the two receivers, so that non-uniform source distributions can introduce amplitude asymmetry or travel-time bias in the recovered EGF (Snieder, 2004; Snieder and Sens-Schönfelder, 2015). Surface-wave phase and group velocities measured from these EGFs can then be mapped spatially and inverted for shear-wave velocity models, as first demonstrated at regional scale by Shapiro and Campillo (2004) and Sabra et al. (2005).

We applied this framework to continuous three-component broadband recordings from the SESAME array spanning 2010–2014, computing cross-correlations for all unique station pairs. We designed the processing to preserve relative amplitudes between vertical and radial components, a requirement for the subsequent extraction of both phase velocities and Rayleigh-wave ellipticity from the cross-correlation functions.

For each day of recording, we loaded all available ambient noise waveforms, verified matching sample rates and synchronized start times between station pairs, and discarded pairs that failed either check. We divided the continuous waveforms into two-hour windows with 50% overlap, detrended and tapered each window with a cosine function, and rotated the horizontal components from geographic coordinates (N, E) into the radial–transverse system aligned with the inter-station azimuth. A key departure from the standard processing workflow of Bensen et al. (2007) is that we did not apply one-bit temporal normalization or conventional spectral whitening, as these techniques discard the amplitude information needed for ellipticity measurements. Instead, we adopted the amplitude-preserving spectral normalization of Lin et al. (2014), in

which we computed a smoothed average power spectrum from the combined vertical and radial components at each station and used it to normalize all three component spectra prior to correlation. This approach broadens the usable bandwidth while preserving the relative vertical-to-radial amplitude ratio essential for measuring Rayleigh-wave ellipticity from the cross-component (ZR and RZ) correlation functions (Lin et al., 2014).

We performed cross-correlation in the frequency domain, forming four correlation functions per window: ZZ (vertical–vertical), RR (radial–radial), ZR (vertical at station 1 with radial at station 2), and RZ (radial at station 1 with vertical at station 2). We normalized each correlation by the smoothed average power spectra of the two stations, accumulated the results into a running coherency sum, and divided by the window count to obtain the daily average coherency. We then transformed this daily average back to the time domain via the inverse Fourier transform and stacked daily correlations into monthly and full-deployment (2010–2014) intervals. The linearity of cross-correlation guarantees that stacking short-window correlations yields the same result as correlating the full-length time series directly (Bensen et al., 2007). We used the full-deployment stacks for dispersion analysis because the signal-to-noise ratio of the recovered EGF improves systematically with stacking duration. To reduce the amplitude asymmetry that arises from non-uniform noise-source distributions, we averaged the causal and acausal sides of each cross-correlation (supplementary material, Figure S3) to form the symmetric component, which we used for all subsequent group and phase velocity measurements.

The Atlantic Coastal Plain provides a favorable noise environment for this analysis. Persistent oceanic microseisms generated by Atlantic wave–wave interactions deliver strong energy in the 0.05–0.3 Hz band from easterly azimuths, populating the stationary-phase end-fire lobes for the predominantly coast-perpendicular SESAME station geometry. Anthropogenic and meteorological sources contribute at higher frequencies from a broader range of directions, extending the usable bandwidth and improving azimuthal coverage. Together these sources yield stable, broadband EGFs across the full range of inter-station distances in the array.

3.3 Dispersion and ellipticity measurements

From the empirical Green's functions obtained in Section 3.2, we extract three complementary observables: Rayleigh-wave phase velocity from cross-spectral Bessel function fitting (Section 3.3.1), group velocity from frequency–time analysis (Section 3.3.2), and Rayleigh-wave ellipticity from multicomponent cross-correlation amplitude ratios (Section 3.5). We then localize the path-averaged dispersion measurements through a tomographic inversion that yields spatially resolved velocity maps and per-station dispersion curves with formal uncertainties (Section 3.3.3). Phase velocity is sensitive primarily to absolute velocity at depth, group velocity to velocity gradients and layer boundaries, and ellipticity to shallow impedance contrasts (Levshin et al., 1989; Bensen et al., 2007; Lin et al., 2014). Measuring all three from the same dataset provides stronger joint constraints on the velocity model than any single observable alone.

3.3.1 Phase velocity from cross-spectral Bessel function fitting

We measured inter-station Rayleigh-wave phase velocities using the cross-spectral formulation of Aki (1957). For a spatially homogeneous noise source distribution and inter-station distances much larger than the wavelength, the real part of the vertical-component (ZZ) cross-spectrum takes the form (Aki, 1957; Cox, 1973):

$$\Re[C_{ZZ}(\omega, r)] = A\, J_0\left(\frac{\omega r}{c_R(\omega)}\right), \tag{13}$$

where $c_R(\omega)$ is the Rayleigh-wave phase velocity at angular frequency ω, r is the inter-station distance, J_0 is the zeroth-order Bessel function of the first kind, and A is an amplitude prefactor. The zero crossings of J_0 strictly encode the phase velocity at each frequency. Following Menke and Jin (2015), we set $A = 1$ because the fit is controlled by the oscillatory structure of the cross-spectrum rather than by its slowly varying amplitude, and we focus exclusively on the fundamental mode, a simplification justified by the tight group-velocity windowing applied during cross-correlation processing, which isolates the fundamental-mode Rayleigh wave from overtone energy (Nishida et al., 2008; Takeo et al., 2013).

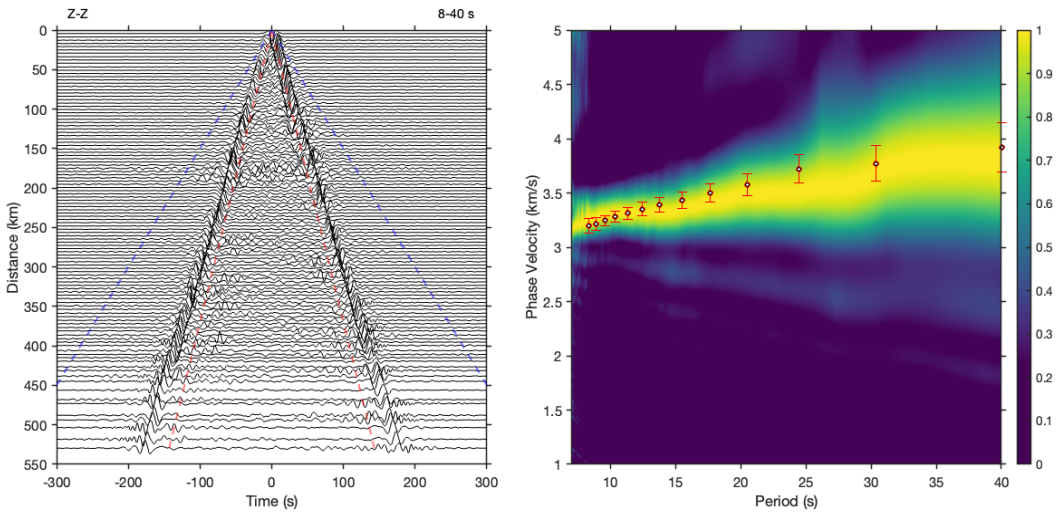

Figure 2: (Left) Stacked vertical-component cross-correlation waveforms sorted by inter-station distance, showing fundamental-mode moveout up to ∼550 km. (Right) Linear Radon Transform velocity–period image with picked fundamental-mode phase velocities (red markers) that serve as the starting model for cross-spectral Bessel function fitting.

Starting model from Linear Radon Transform: Fitting Bessel functions to observed cross-spectra requires a reliable starting model to avoid the cycle-skipping ambiguities inherent to the oscillatory nature of J_0. We established this baseline using the Linear Radon Transform (LRT) (Schultz and Gu, 2012; An et al., 2007), which maps the ensemble of stacked cross-correlation waveforms from the time–distance domain into the slowness–frequency domain, producing a two-dimensional image where surface-wave modes emerge as coherent energy bands. We computed the LRT via a conjugate gradient inversion with frequency-dependent weighting that promotes sparsity and enhances mode-branch resolution (Ji, 2006)

Figure 2 shows the stacked vertical-component cross-correlation waveforms sorted by inter-station distance (left) and the corresponding LRT velocity–period image (right). The fundamental-mode dispersion ridge is clearly resolved as a continuous band extending from ∼3.2 km/s at short periods to ∼3.9 km/s at 35 s. Below 8 s the fundamental mode becomes difficult to separate from scattered energy, and above 35 s coherent energy diminishes, so we restricted our LRT picks to the 8–35 s band. The resulting dispersion curve, interpolated to 40 evenly spaced frequency points, served as the starting phase velocity model $c_{\mathrm{start}}(\omega)$ for individual station-pair fitting.

Bessel function fitting procedure: We extracted inter-station phase velocities by adapting the methodology of Menke and Jin (2015) to fit Eq. (13) to the observed real part of the ZZ cross-spectrum. Prior to inversion, we applied a group-velocity window to the cross-spectrum to isolate fundamental-mode energy and suppress noise outside the signal band (supplementary material, Figure S1). We formulated the phase velocity extraction as a nonlinear least-squares problem in which travel time $t(\omega) = r/c(\omega)$ at each discrete frequency is the model parameter, and to mitigate cycle skipping, we performed the inversion in two passes. In the first pass, the LRT-derived curve is the starting model and residuals are weighted by inverse angular frequency, emphasizing longer periods where cycle-skipping risk is greatest. We then averaged the resulting curves from all station pairs, weighted by individual misfits, to produce a network-averaged dispersion curve. This refined curve served as the starting model for the second pass, which uses uniform frequency weighting. This two-pass strategy effectively suppresses cycle-skipping errors without requiring an exhaustive grid search over starting models.

Figure 3 illustrates the workflow for station pair D02–W02 with the best-fitting J_0 function accurately tracking the observed cross-spectral oscillations across the 0.03–0.12 Hz bandwidth, and the resulting phase velocities decreasing from ∼3.9 km/s at 33 s to ∼3.2 km/s at 9 s, which is consistent with the transition

from deeper crystalline basement to shallow sedimentary velocities.

Quality control and error estimation: We imposed a minimum wavelength criterion requiring the inter-station distance to exceed $N_{\mathrm{wl}} = 2$ wavelengths at each measurement period:

$$\frac{r}{T\,c(T)} > N_{\mathrm{wl}}, \tag{14}$$

where T is the period, and we excluded station pairs failing this criterion at a given period. We derived uncertainties from the model covariance matrix of the nonlinear least-squares fit; the full formulation is given in Appendix A.

3.3.2 Group velocity from frequency–time analysis

We measured group velocity dispersion using frequency–time analysis (FTAN), following the automated framework of Bensen et al. (2007) with modifications to the ridge-tracking algorithm applied to the SESAME array (Dziewonski et al., 1969; Levshin et al., 1989; Ritzwoller and Levshin, 1998). We filtered the symmetric cross-correlation function for each station pair with a suite of 50 narrow-band Gaussian filters with center frequencies linearly spaced from 1/25 to 1/3 Hz (25 to 3 s period) and a constant $1/e$ half-width (the filter definition is given in Appendix A), transformed each narrow-band trace back to the time domain, and obtained the envelope function $|A(t, \omega_0)|$ via the Hilbert transform. Collecting envelopes across all center frequencies produces a two-dimensional frequency–time diagram in which surface-wave energy appears as a coherent ridge whose time position varies with period according to the group velocity dispersion.

At each center frequency, we identified the group arrival time $\tau(\omega_0)$ as the envelope maximum within the window $t \in [r/U_{\max}, r/U_{\min}]$, where $U_{\min} = 1.0$ km/s and $U_{\max} = 6.0$ km/s, so that the group velocity follows as:

$$U(\omega_0) = \frac{r}{\tau(\omega_0)}. \tag{15}$$

We then constructed a continuous dispersion curve from the discrete envelope peaks using an automated ridge-tracking algorithm that optimizes a combined score of envelope amplitude and curve smoothness, enforcing a smoothness penalty ($\lambda_{dU} = 30$) and a maximum velocity jump of 0.25 km/s per period step.

Multi-component FTAN: We applied FTAN independently to all four cross-correlation components (RR, RZ, ZR, ZZ). Figure 4 illustrates the resulting frequency–time diagrams for station pair D02–W02 (368 km). All four components exhibit a coherent energy ridge increasing from ∼2.0 km/s at 3–5 s to ∼3.0 km/s at 15–25 s, consistent with increasing depth sensitivity from shallow sediments to crystalline basement. The cross-components (RZ and ZR) track the same dispersion branch as the auto-components (RR and ZZ), confirming that the amplitude-preserving normalization of Section 3.2 retains coherent inter-component phase information. We also applied FTAN independently to the causal and acausal signals before symmetric averaging; consistency between the two sides, quantified through correlation coefficients among the causal, acausal, and symmetric dispersion curves (supplementary material, Figure S3), provides an internal check on noise-source illumination quality.

Quality control: We retained group velocity measurements only if two criteria were satisfied. First, the inter-station distance must exceed three wavelengths at each measurement period ($\lambda = U \cdot T$). Second, the FTAN-envelope signal-to-noise ratio must exceed 20 at the measurement period, computed as the ratio of peak envelope amplitude within the signal window to the root-mean-square amplitude in a trailing noise window; this threshold is more conservative than the value of 10 recommended by Bensen et al. (2007) but is warranted by the absence of multi-year seasonal averaging for independent uncertainty estimation. We evaluated both criteria independently at each period for the causal and acausal sides, enabling frequency-dependent data selection.

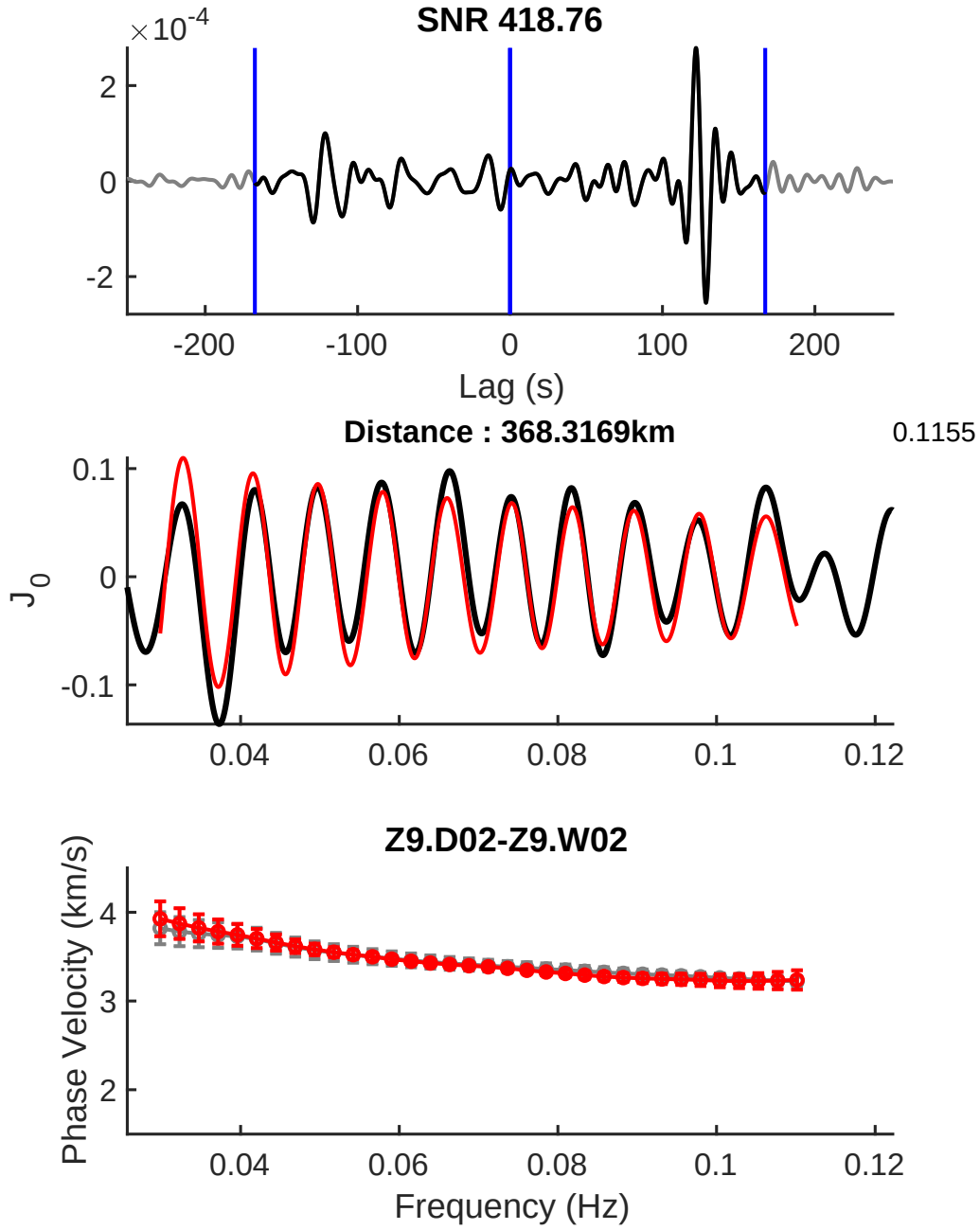

Figure 3: Cross-spectral Bessel function fitting for station pair Z9.D02–Z9.W02 (∼368 km). (Top) Time-domain cross-correlation function (SNR = 419). Blue lines denote fundamental-mode windowing. (Middle) Observed real part of the ZZ cross-spectrum (black) and best-fitting J_0 function (red). (Bottom) Extracted phase velocity (red) compared with the starting model (gray).

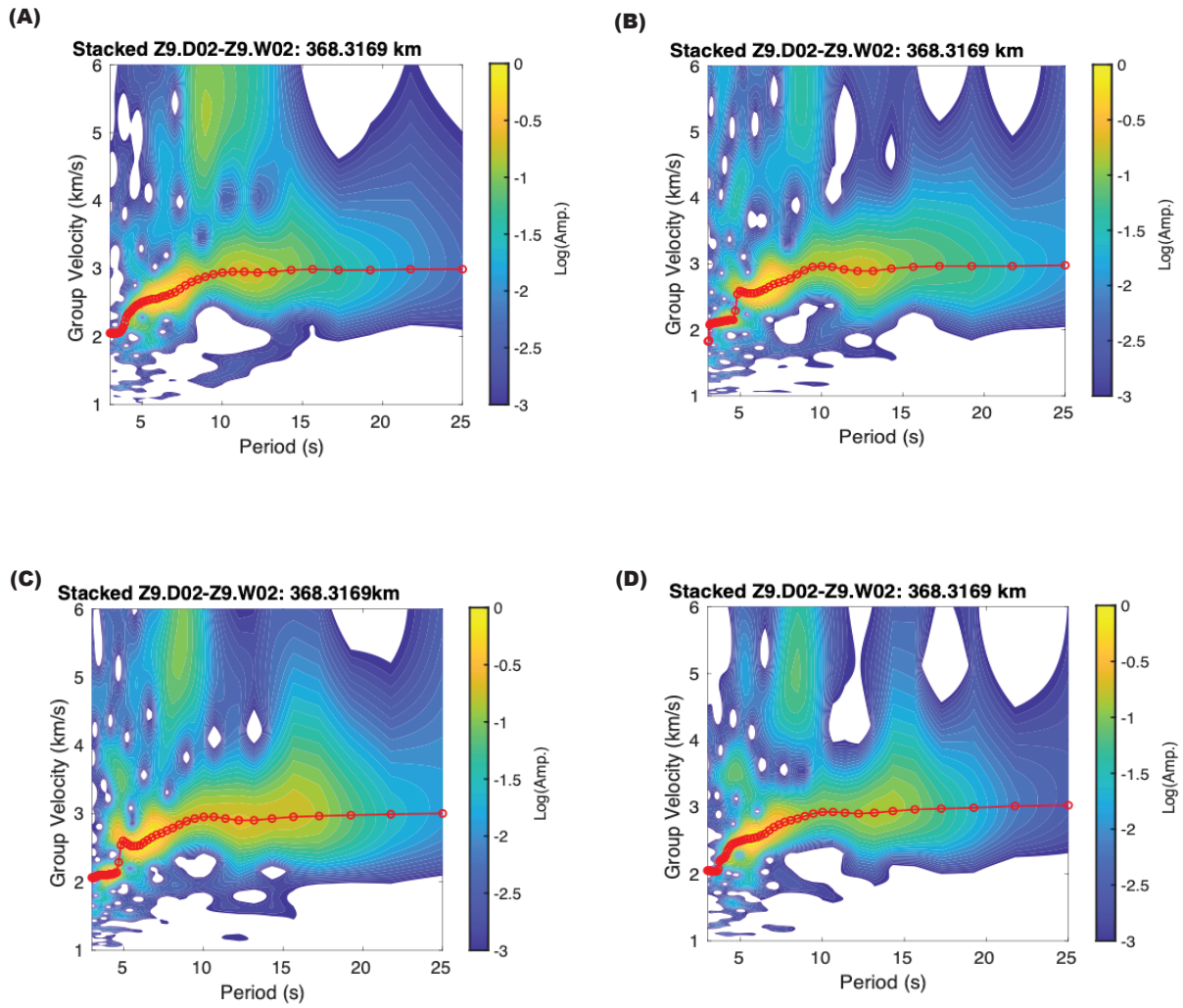

Figure 4: FTAN diagrams for station pair Z9.D02–Z9.W02 (368 km). Panels show (A) RR, (B) RZ, (C) ZR, and (D) ZZ causal and acausal stacked components. Color scale indicates log-amplitude of the narrow-band filtered envelope. Red circles mark automatically picked group velocities.

3.3.3 Tomographic inversion of dispersion measurements

The inter-station phase and group velocity measurements described above are path-averaged quantities that integrate the velocity structure along each station pair. To localize these measurements and obtain spatially resolved velocity maps from which per-station dispersion curves can be extracted, we performed a two-dimensional ray-theory tomographic inversion following Jin and Gaherty (2015), solving simultaneously for isotropic velocity and 2θ azimuthal anisotropy on a regular geographic grid by parameterizing the Rayleigh-wave phase (or group) velocity at propagation azimuth θ and period T as:

$$c(\theta, T) = c_0(T) + A_{c2}(T)\cos(2\theta) + A_{s2}(T)\sin(2\theta), \tag{16}$$

where c_0 is the isotropic velocity and A_{c2} and A_{s2} capture the 2θ azimuthal variation. We neglected higher-order 4θ terms because their sensitivity is negligibly small for fundamental-mode Rayleigh waves (Smith and Dahlen, 1973; Montagner and Tanimoto, 1991), and although we solved for the anisotropic terms to prevent azimuthally unresolved structure from leaking into the isotropic maps, we focus exclusively on the isotropic component $c_0(T)$ in this study.

Because the SESAME array comprises three approximately linear sub-arrays with distinct station spacing and azimuthal coverage, we performed the tomographic inversion separately for each transect (D, E, and W) on grids tailored to its geographic extent, excluding station pairs that cross between sub-arrays; this per-line strategy keeps the regularization and ray-density masking appropriate to each geometry and avoids spurious structure from long, oblique inter-line paths. We solved for the isotropic field on a 0.1° grid and for the anisotropy parameters on a coarser 0.5° grid reflecting the longer spatial wavelength over which anisotropy can be resolved. For phase velocity, we inverted inter-station traveltimes from Bessel-function fitting of the ZZ cross-spectra (Section 3.3.1) across the 9–34 s band, requiring a minimum signal-to-noise ratio of 5, a maximum inter-station distance of 750 km, and a minimum separation of two wavelengths at each period. For group velocity, we used the ZZ component (Section 3.3.2) for consistency with the phase dataset and inverted traveltimes across the 3–25 s band, imposing a stricter SNR threshold of 20 to compensate for the greater scatter in envelope-based arrival-time picks, together with a minimum causal–acausal correlation coefficient of 0.8 so that only measurements with consistent dispersion in both propagation directions were retained.

We formulated the inverse problem as a damped least-squares inversion for slowness, with second-derivative smoothing on the isotropic field, norm damping on the anisotropic parameters where ray coverage is sparse, and separate smoothing on the anisotropic fields; the augmented system, data weighting scheme, and outlier rejection procedure are given in Appendix B. We estimated model uncertainties at each grid node from the diagonal of the posterior model covariance matrix and characterized lateral resolving power through checkerboard tests at 0.5°, 1.0°, and 1.5° scales, which demonstrate faithful recovery at scales of 1.0° and larger across the central portions of each sub-array, with degradation toward the array periphery, so features smaller than ∼1° should be interpreted with caution. To obtain the per-station dispersion curves used in the remainder of this study, we extracted the isotropic velocity and its tomographic standard deviation at each station from the nearest resolved grid node, requiring the station to lie within two grid cells (∼22 km at 0.1° spacing) of a valid node and assigning missing values at periods where this threshold was exceeded.

3.4 HVSR processing and quality control

We computed HVSR from continuous ambient-noise records using hvsrpy (Vantassel, 2025), which implements the diffuse-field processing framework consistent with the theoretical basis of Section 3.1. Processing follows the SESAME recommendations (Bard and SESAME participants, 2005) and proceeds through hourly HVSR computation and statistical quality control with our parameters summarized in Table 1.

We processed hourly MiniSEED data for all 88 SESAME stations over 31 consecutive days of recording (744 hourly records per station) to assess temporal variability in the region. The daily mean HVSR curves proved stable throughout this interval at every station, consistent with the expected temporal stability of microtremor HVSR, so we based our station-level compilation on a single 24-hour recording once anomalous hours were removed as described below. Channels ending in "1" or "2" were remapped to "N" and "E" to ensure consistent three-component identification. Within each hourly record, we demeaned, detrended, corrected for instrument response, subdivided the record into 300-s windows, and tapered each window with a Tukey

Table 1: HVSR processing parameters.

Parameter	Value
Processing framework	Diffuse field assumption (DFA)
Input components	E, N, Z (three-component)
Assessment period per station	31 days (744 hourly records)
Analysis period per station	1 day (24 hourly records)
Data segmentation	300-s windows within each 1-hour record
Preprocessing	Mean removal, linear detrend
Instrument response	Corrected to ground motion
Tapering	Tukey window ($\alpha = 0.1$)
Spectral smoothing	Log-rectangular, bandwidth 0.1
Frequency range	0.2 Hz to Nyquist
Frequency samples	256 (log-spaced)
Anti-triggering	STA/LTA ($0.2 < \text{STA/LTA} < 2.5$)
QC rejection threshold	$\bar{\chi}^2_h > 3$

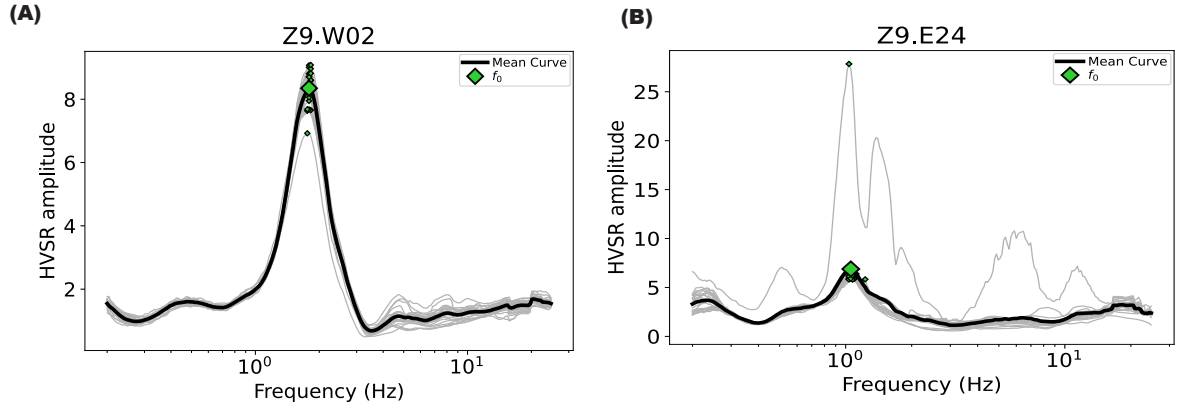

Figure 5: Hourly HVSR curves over 24 hours at two SESAME stations exhibiting varying characters. (a) W02: stable curves with a consistent peak near 2 Hz. (b) E24: one anomalous hour produces a spurious high-amplitude peak.

window ($\alpha = 0.1$). We computed Fourier amplitude spectra for the vertical and two horizontal components at 256 logarithmically spaced frequencies between 0.2 Hz and the Nyquist frequency, smoothed using a log-rectangular operator, and applied an STA/LTA anti-triggering criterion to suppress short transient bursts within the windows. We restricted the HVSR band to frequencies above 0.2 Hz to avoid modal contamination at lower frequencies; the ellipticity measurements of Section 3.5 carry the site characterization below this limit.

For each day (24 hourly records), we computed the mean and standard deviation of the HVSR across frequencies:

$$\overline{\text{HVSR}}(f) = \frac{1}{24}\sum_{h=1}^{24} \text{HVSR}_h(f), \qquad \sigma_{\text{HVSR}}(f) = \sqrt{\frac{1}{23}\sum_{h=1}^{24}\left[\text{HVSR}_h(f) - \overline{\text{HVSR}}(f)\right]^2}. \tag{17}$$

After visually inspecting the resulting HVSR curves, we found inconsistencies at some stations where anomalous hourly records shift the daily mean and inflate the standard deviation, obscuring the true site resonance. The STA/LTA criterion suppresses short bursts within a 300-s window but passes disturbances that persist for most of an hour, so we introduced a quality control scheme that removes these spurious hours without

affecting the hours that are stable. We defined a reduced chi-square misfit that quantifies how far each hourly HVSR deviates from the daily mean relative to the observed variability:

$$\chi_h^2(f) = \frac{\left[\mathrm{HVSR}_h(f) - \overline{\mathrm{HVSR}}(f)\right]^2}{\sigma_{\mathrm{HVSR}}^2(f)}. \tag{18}$$

Averaging over all 256 frequency samples yields a single scalar per hourly record:

$$\bar{\chi}_h^2 = \frac{1}{256}\sum_{i=1}^{256} \chi_h^2(f_i). \tag{19}$$

We rejected hours with $\bar{\chi}_h^2 > 3$, a deliberately generous threshold corresponding approximately to deviations exceeding two standard deviations when averaged across frequency; genuine hour-to-hour site variability falls comfortably below this value, while spurious hours exceed it by a wide margin. We then recomputed the daily mean and standard deviation from the retained hours and repeated the procedure independently for each of the 31 recording days at every station. Across the array, this scheme typically rejected between zero and two hours per station-day.

We then applied the quality control procedure to HVSR results from all 88 SESAME stations, and we found that at many locations the hourly HVSR curves exhibit highly consistent peak shapes throughout the entire 24-hour recording period (Figure 5a). Station W02 illustrates this ideal case: all 24 hourly curves cluster tightly around the mean, producing a clean single peak near 2 Hz with amplitude ∼9 and stable f_0 picks across hours. When we applied the reduced chi-square metric, every hour yielded a low misfit ($\bar{\chi}_h^2$ ranging from 0.38 to 2.40, all below the rejection threshold of 3), so no hours were rejected (Figure 6a,b). The recomputed mean curve and the mean $\pm 3\sigma$ envelope confirm that the site response is well constrained and the daily compilation represents the true site resonance.

Conversely, at other stations the initial 24-hour compilation reveals anomalous hours that severely distort the overall mean. Station E24 illustrates this clearly: the majority of hourly curves show a consistent double-peaked site response near 0.6 Hz and 1.5 Hz, but a single spurious hour, likely caused by a localized transient disturbance or equipment glitch, produces an uncharacteristically high-amplitude peak reaching ∼27 near 1.2 Hz that masks the true fundamental frequency (Figure 5b). When we applied the reduced chi-square metric, this anomalous hour (Hour 20) generated an exceptionally high misfit of 16.04, far exceeding the threshold of 3, while all other hours remained well below it with misfits ranging from 0.11 to 1.26 (Figure 6c). The algorithm successfully flagged and removed this spurious hour, and as shown in Figure 6d, recomputing the mean and standard deviation from the retained hours recovered the true f_0 and preserved the natural amplitude of the site response, entirely free from the bias introduced by the anomalous hour. The mean $\pm 3\sigma$ envelope in the recomputed result shows tight clustering of all retained hours, confirming that the quality control procedure isolates and removes only the genuinely anomalous data.

We measured the fundamental peak frequency f_0 at each station from the final quality-controlled mean HVSR curve as the frequency of maximum H/V amplitude where a clear peak is present and estimated uncertainty in f_0 from the peak bandwidth and the inter-hour variability of peak position among the retained hours, following the reliability criteria of Bard and SESAME participants (2005). To obtain a robust station-level estimate, we averaged the predominant frequency and its amplitude from each retained hourly record across the full 24-hour period. This station-level compilation across all 88 SESAME stations provides the fundamental frequency distribution (Figure 8) used for regional characterization of the Atlantic Coastal Plain sedimentary structure.

3.5 Rayleigh-wave ellipticity from multicomponent cross-correlations

We measured fundamental-mode Rayleigh-wave ellipticity following the approach of Lin et al. (2014), which exploits the fact that ratios among the four cross-correlation components (ZZ, RR, ZR, RZ) yield estimates of the H/V ratio at each station in a pair without requiring single-station recordings.

Using the FTAN procedure described in Section 3.3.2, we extracted the peak envelope amplitude at the fundamental-mode group arrival time for the ZZ, ZR, RZ, and RR components on both the causal and acausal sides of each correlogram. For the first station in each pair, we computed H/V from the RZ/ZZ and

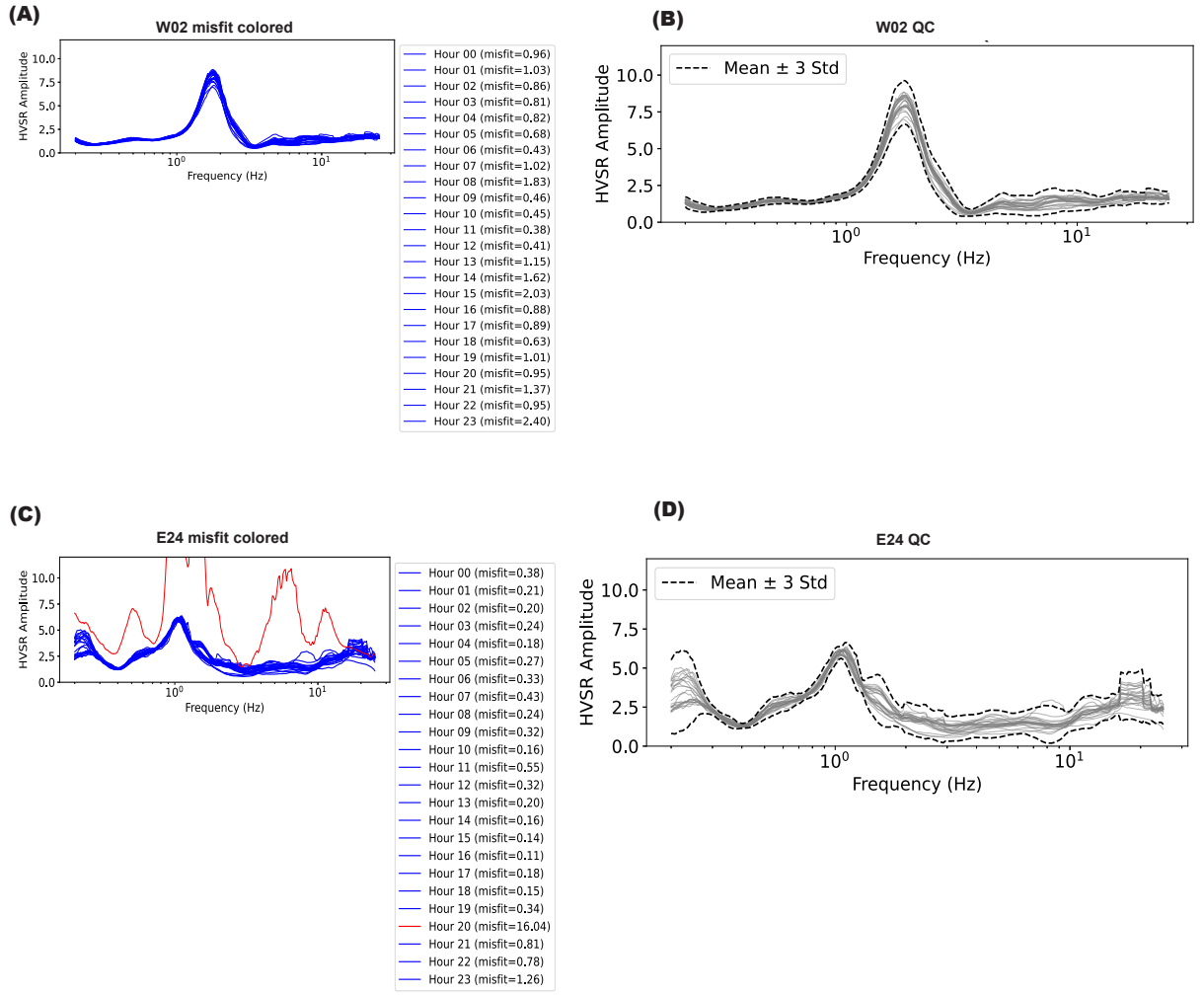

Figure 6: Reduced chi-square quality control applied to the two stations of Figure 5. (a, b) W02: all hours pass ($\bar{\chi}^2_h < 3$). (c, d) E24: the anomalous hour (red) is rejected and the recomputed mean recovers the true site response.

RR/ZR amplitude ratios; for the second station, from ZR/ZZ and RR/RZ. Averaging the two complementary estimates at each station yields a single ellipticity value per period.

To verify that the measured ratios correspond to genuine fundamental-mode Rayleigh-wave polarization, we computed the instantaneous phase difference between vertical and radial components at the group arrival time. We adopted the convention that $-90°$ indicates retrograde particle motion (Figure 7a), the expected sense for fundamental-mode Rayleigh waves in a normal velocity structure, and corrected for the radial-direction sign convention at station 1.

We retained only measurements that passed several selection criteria at each period. First, we required the minimum signal-to-noise ratio across all four components (ZZ, RR, ZR, RZ) to exceed 8, computed as the ratio of peak envelope amplitude within the expected Rayleigh-wave signal window to the root-mean-square noise level outside this window; this threshold is lower than the value of 20 used for group velocity because requiring it simultaneously on all four components is itself highly selective. Second, we required the inter-station distance to exceed three wavelengths at the measurement period to satisfy the far-field condition (Bensen et al., 2007). Third, we retained only measurements with instantaneous phase differences within $-90° \pm 45°$, confirming fundamental-mode Rayleigh-wave particle motion (Figure 7b,c) and rejecting measurements that may be contaminated by overtones or non-Rayleigh energy. Fourth, we required the group arrival times measured on the RR, RZ, and ZR components to agree with the ZZ arrival time to within five times the measurement period, a tolerance that scales with the natural broadening of the group envelope at longer periods and ensures that all components sample the same surface-wave energy. Finally, we required the fractional difference between the two independent H/V estimates at each station not to exceed 10%, providing an internal consistency check on the fundamental-mode character of the measurement, and we discarded measurements that failed any of these criteria at a given period.

We applied this method to all cross-correlations to establish a large number of H/V measurements for each station across the region and further stabilized each station's result by removing spurious measurements that exceeded three standard deviations from the mean. For each station, we then computed isotropic H/V as the median across all accepted measurements from contributing station pairs, with uncertainty given by the standard deviation. This averaging over multiple independent paths suppresses path-specific biases and produces a robust, locally representative ellipticity curve at each station. Because H/V is a measurement of the relative change of two quantities, we expressed both H/V and uncertainty as logarithms; this has been shown to be the only symmetric, additive, and normed indicator of relative change (Törnqvist et al., 1985; Berg et al., 2018).

Figure 7 illustrates the procedure for a representative station pair consisting of a Blue Ridge site on crystalline rock and a Coastal Plain site overlying thick sediments. Panel (A) shows cross-correlation waveforms at 15.6 s period alongside hodograms that confirm the expected elliptical particle motion: the mountain station exhibits compact ellipses with H/V < 1 (vertical motion dominant), while the coastal station shows elongated ellipses with H/V > 1, reflecting amplified horizontal motion in the low-velocity sedimentary column. Panels (B) and (C) display the resulting ellipticity curves, phase-difference diagnostics, and H/V histograms at selected periods for the two stations. The mountain station yields stable H/V values of 0.6–0.9 across the 4–24 s band, whereas the coastal station shows a pronounced increase from $\sim$0.8 at long periods to 2–3 below 6 s, consistent with the strong sediment–basement impedance contrast. Phase differences at both stations cluster tightly near $-90°$, confirming retrograde fundamental-mode Rayleigh-wave polarization across the measurement band and ruling out overtone contamination, which would appear as prograde motion.

4 Results

4.1 HVSR

The HVSR results across the SESAME array (Figure 8) reveal spatial patterns that are strongly correlated with the geological structure of the southeastern Atlantic Coastal Plain. Across the array, HVSR peak frequencies span more than an order of magnitude, from $<$0.5 Hz at stations in southern Georgia and along the Georgia–Florida coastal margin to $\sim$5–15 Hz at stations nearer the Fall Line. The systematic decrease in f_0 toward the southeast is consistent with the quarter-wavelength scaling between resonance frequency and sediment thickness ($f_0 \approx V_S/4H$): the spatial gradient in f_0 broadly parallels the sediment-thickness

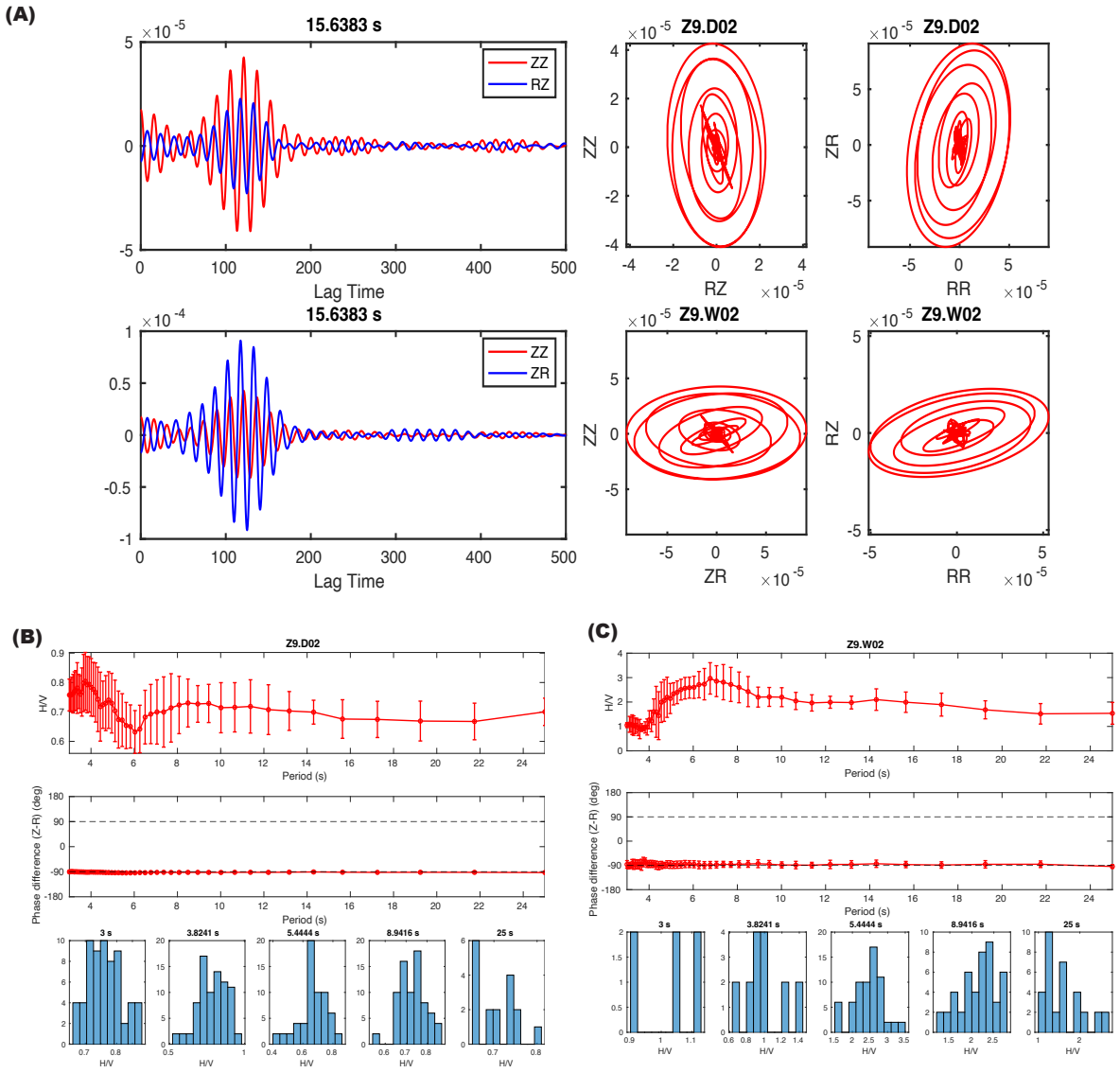

Figure 7: Ellipticity measurement for station pair Z9.D02–Z9.W02. (A) Cross-correlation waveforms at 15.6 s with hodograms: Z9.D02 (Blue Ridge) shows compact ellipses (H/V $<$ 1); Z9.W02 (Coastal Plain) shows elongated ellipses (H/V $>$ 1). (B) Results for Z9.D02: median H/V curve, phase-difference diagnostic, and H/V histograms. (C) Same for Z9.W02, where H/V increases from $\sim$0.8 at long periods to 2–3 below 6 s.

contours of Boyd et al. (2024), with the most abrupt transition occurring where mapped thickness increases from <0.5 km near the Piedmont margin to >2 km in the southern embayment.

Along the eastern transects, f_0 varies coherently with regional basement depth. Stations on the South Carolina transect show a west-to-east decrease in f_0 as the line crosses from the Piedmont into the inner Coastal Plain, tracking the ∼0.5–1 km thickness contours. The two north–south transects in eastern Georgia show an analogous progression, transitioning from $f_0 \approx$ 3–5 Hz in the north to <0.5 Hz in the south as sediment thickness increases from <0.5 km to >2 km. Near the southern end of the eastern array, isolated higher-frequency peaks coincide with locally reduced thickness (∼1 km), suggesting that spatial variations in basement relief imprint directly on the observed resonance frequency. Taken together, these patterns indicate that, across much of the eastern Atlantic Coastal Plain, the primary impedance contrast controlling HVSR is closely tied to the sediment–basement interface.

In contrast, the western array exhibits departures from the simple thickness–frequency trend we observed in the east. Near the southwestern portion of the array, we observe relatively high resonance frequencies (approaching ∼10 Hz) even where regional contours indicate a persistently thick sedimentary column (∼1.5–2 km), and we interpret this as evidence that HVSR in this region responds to a shallower impedance contrast within the sedimentary column, potentially produced by highly compacted or lithified units at depth that act as the effective seismic basement above the true crystalline basement (Section 5.1). Alternatively, or in combination, higher average shear-wave velocities within the western sediment package could elevate f_0 for a given thickness through the $V_S/4H$ relationship. This east–west divergence implies that a single f_0–thickness calibration may not be transferable across the Atlantic-to-Gulf Coastal Plain transition, and it motivates a joint-inversion parameterization that allows laterally varying sediment properties rather than assuming a uniform clastic column. At several stations in the central array, secondary higher-frequency peaks are also present, which we interpret as resonances associated with internal impedance contrasts within the layered sedimentary succession; this behavior is qualitatively consistent with patterns in our ellipticity measurements (Section 4.2).

4.2 Rayleigh-wave ellipticity

The spatial distribution of fundamental-mode Rayleigh-wave ellipticity across the SESAME array reveals clear, period-dependent patterns that directly reflect the depth-varying sensitivity of particle motion to underlying geological structure (Figure 9). At 4 s (Figure 9a), the wavefield exhibits high sensitivity to the upper few kilometers, producing the strongest spatial variation we observe across the array. Stations on the crystalline rock of the Blue Ridge and Piedmont exhibit low H/V ratios ($\log_{10}$(H/V) ≈ −0.2 to −0.4), indicating the near-vertical particle motion expected where thin or absent sedimentary cover offers minimal acoustic impedance contrast, consistent with the low HVSR amplitudes at these stations (Figure 8). Transitioning into the Coastal Plain, the H/V ratio increases dramatically; the highest ellipticities ($\log_{10}$(H/V) ≈ 0.6 to 1.0, or H/V ≈ 4 to 10) occur along both transects in central Georgia near 31–32°N, where sediment thickness ranges from 1–2 km. Crucially, these peak values do not coincide with the thickest sedimentary packages. Instead, they occur at intermediate basin depths where the 4 s Rayleigh wave's depth sensitivity window optimally straddles the sediment–basement interface, sampling the sharp impedance contrast most directly.

Further south, where sediments exceed 3 km, the 4 s waves become fully trapped within the thick sedimentary column and do not sense the deeper crystalline basement, causing H/V values to moderate ($\log_{10}$(H/V) ≈ 0.1 to 0.3). Along the eastern South Carolina transect, a clear west-to-east progression from low to elevated H/V tracks the thickening sedimentary wedge, mirroring the f_0 gradient described in our HVSR results. At 17 s (Figure 9b), the wavefield penetrates to deeper upper-crustal structure, and the spatial ellipticity pattern becomes substantially more uniform. The overall range of H/V values narrows noticeably ($\log_{10}$(H/V) ≈ −0.4 to 0.5, compared with −0.4 to 1.0 at 4 s) and the anomalies observed at 4 s in the central Coastal Plain diminish as the longer-period Rayleigh waves average over a much deeper lithological column, losing sensitivity to the shallow, high-contrast interfaces that dominate the short-period response. However, stations overlying the thickest sediments in the deep southern embayment retain slightly elevated H/V ratios relative to the northern stations, indicating that 17 s waves are beginning to sense the deep sediment–basement interface in those regions.

The associated ellipticity uncertainties (Figure 9c,d) show a general decrease at longer periods. At 4 s, the majority of stations exhibit low uncertainties ($\sigma \approx$ 0.02 to 0.05), but a distinct cluster of elevated

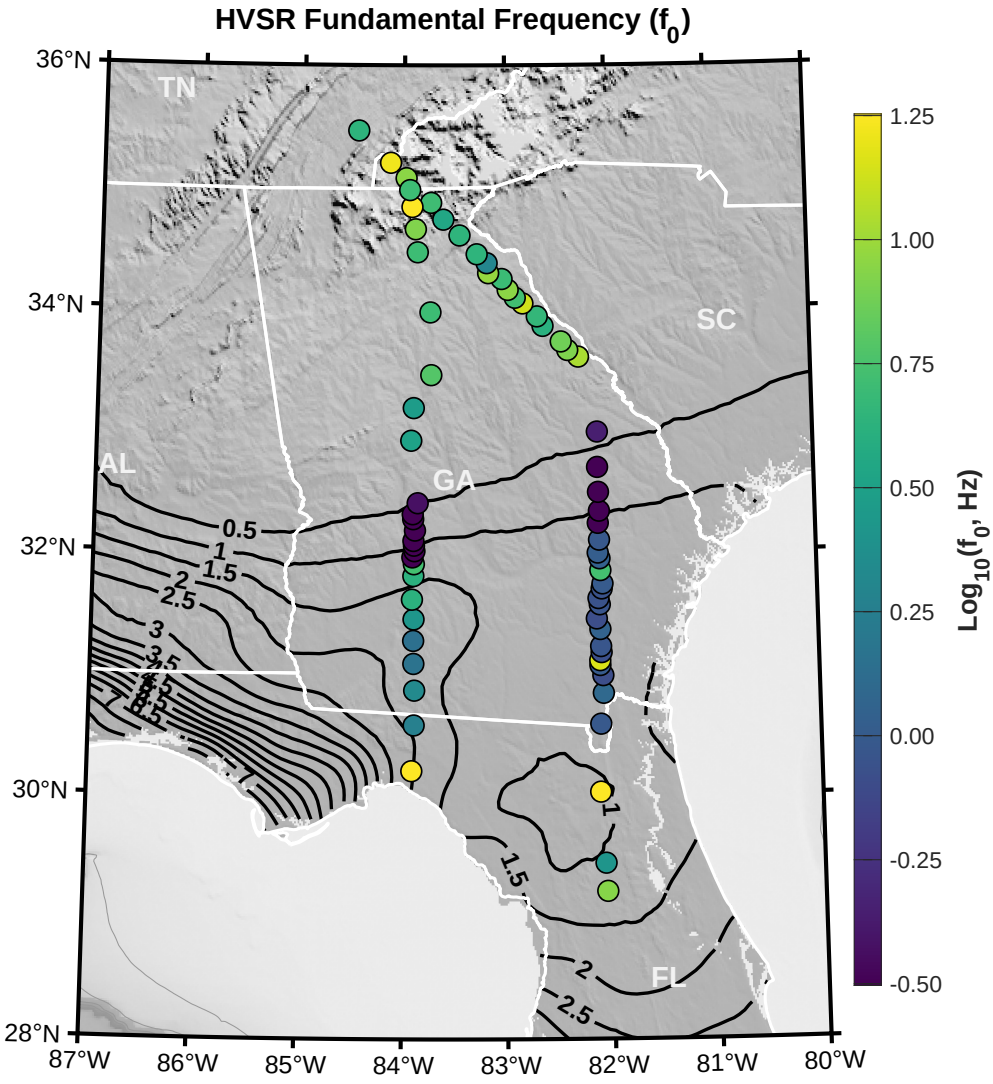

Figure 8: Map of HVSR fundamental resonance frequency f_0 at each SESAME station ($\log_{10}$ color scale) with sediment-thickness contours (km) from Boyd et al. (2024).

uncertainty ($\sigma \approx 0.2$ to 0.4) emerges along the western transect in central Georgia. This localized zone coincides with the region where sediment thickness rapidly increases from 1 to 2.5 km, and we attribute the elevated short-period uncertainty to wavefield complexity, such as multipathing and scattering, generated by severe lateral heterogeneity and steep structural gradients within this section of the sedimentary wedge. At 17 s, uncertainties decrease significantly, with maxima capping around 0.20 and most stations falling well below 0.05. The few persisting elevated uncertainties along the eastern transect and near the Fall Line likely stem from reduced azimuthal coverage at the array margins rather than structural complexity.

These period-dependent spatial patterns carry critical implications for structural modeling. The 4 s ellipticity measurements provide exceptionally strong constraints on the impedance contrast at the sediment–basement interface for intermediate basin depths (1–2 km), while the 17 s data anchor the deeper crustal structure. At the deepest southern stations, where short-period ellipticity samples only the internal sedimentary column, the HVSR data (Figure 8) successfully isolate the fundamental resonance frequency and thus total thickness.

4.3 Phase and group velocity dispersion

The transition from crystalline Piedmont basement to the thick sedimentary wedge of the Atlantic Coastal Plain produces a first-order signature in the Rayleigh-wave dispersion that is clearly resolved by the SESAME array. Figure 10 maps phase velocity at 9, 17, and 34 s (panels a–c) and group velocity at 4, 9, and 17 s (panels d–f) at each station alongside sediment-thickness contours. At all periods, low-velocity regions correspond to the Coastal Plain sedimentary section, while high-velocity regions correspond to the Piedmont and southern Appalachians, and the magnitude of this contrast diminishes systematically with increasing period as Rayleigh-wave sensitivity shifts from shallow sedimentary to deeper crustal structure (supplementary material, Figures S8–S10). These patterns are broadly consistent with existing continental-scale phase and group velocity maps (Ekström, 2017; Herrmann et al., 2021; Shen and Ritzwoller, 2016), but the closer station spacing of the SESAME array (∼9–50 km versus 70–200 km effective resolution in national models) resolves the sharp velocity gradients across the Fall Line and within the sedimentary wedge that are smoothed in continental compilations.

At 9 s (panel a), we observe phase velocities of approximately 2.9–3.05 km/s on the Coastal Plain where sediments are thickest, increasing to 3.3–3.4 km/s at northern stations on the Blue Ridge and Piedmont. The north–south gradient closely parallels the sediment-thickness contours, with a few stations near 30°N on the western transect reaching the lowest phase velocities (∼2.9 km/s) coinciding with the thickest portions of the sedimentary column. The total velocity contrast of approximately 0.3–0.4 km/s across the Fall Line is consistent with the low-velocity corridor identified earlier by Herrmann et al. (2021) and Ekström (2017) at comparable periods. At 17 s (panel b), the spatial pattern persists but the velocity range narrows to approximately 3.3–3.6 km/s as longer-period Rayleigh waves begin sampling below the sedimentary section into the upper crystalline crust, and by 34 s (panel c), phase velocities increase substantially to approximately 3.7–4.3 km/s, and the character of the spatial pattern changes: the highest velocities (∼4.1–4.3 km/s) occur at northern Blue Ridge stations while Coastal Plain stations show lower values (∼3.8–3.9 km/s). At this period, Rayleigh waves are sensitive to the full crustal column and the uppermost mantle, so the pattern reflects crustal thickness variations and deep lithospheric structure rather than the shallow sedimentary section. The progressive convergence from 9 s to 17 s and the transition to crustal-scale sensitivity at 34 s are consistent with the depth-dependent behavior documented by Shen and Ritzwoller (2016) and Yang et al. (2021).

Group velocities reveal an even stronger sedimentary signature than phase velocities at equivalent periods, as expected from their greater sensitivity to shallow structure. At 4 s (panel d), we observe the most obvious lateral contrast in the entire dataset, a north–south gradient spanning approximately 1.9–3.2 km/s, the largest velocity range of any of the observable period. Northern Blue Ridge stations reach ∼3.0–3.2 km/s while stations in the southern portion of the array where sediment thickness exceeds 3 km drop to ∼2.0–2.2 km/s. This contrast exceeds 1 km/s and reflects the extreme sensitivity of short-period group velocity to the low-velocity Cretaceous-to-Cenozoic sedimentary fill (Gann-Phillips et al., 2024; Boyd et al., 2024). We note that the western transect shows slightly higher group velocities at the southern end compared with the eastern transect at equivalent latitudes, a pattern that may reflect the same east–west structural difference we identified in the HVSR f_0 results, where the western sedimentary column appears to contain

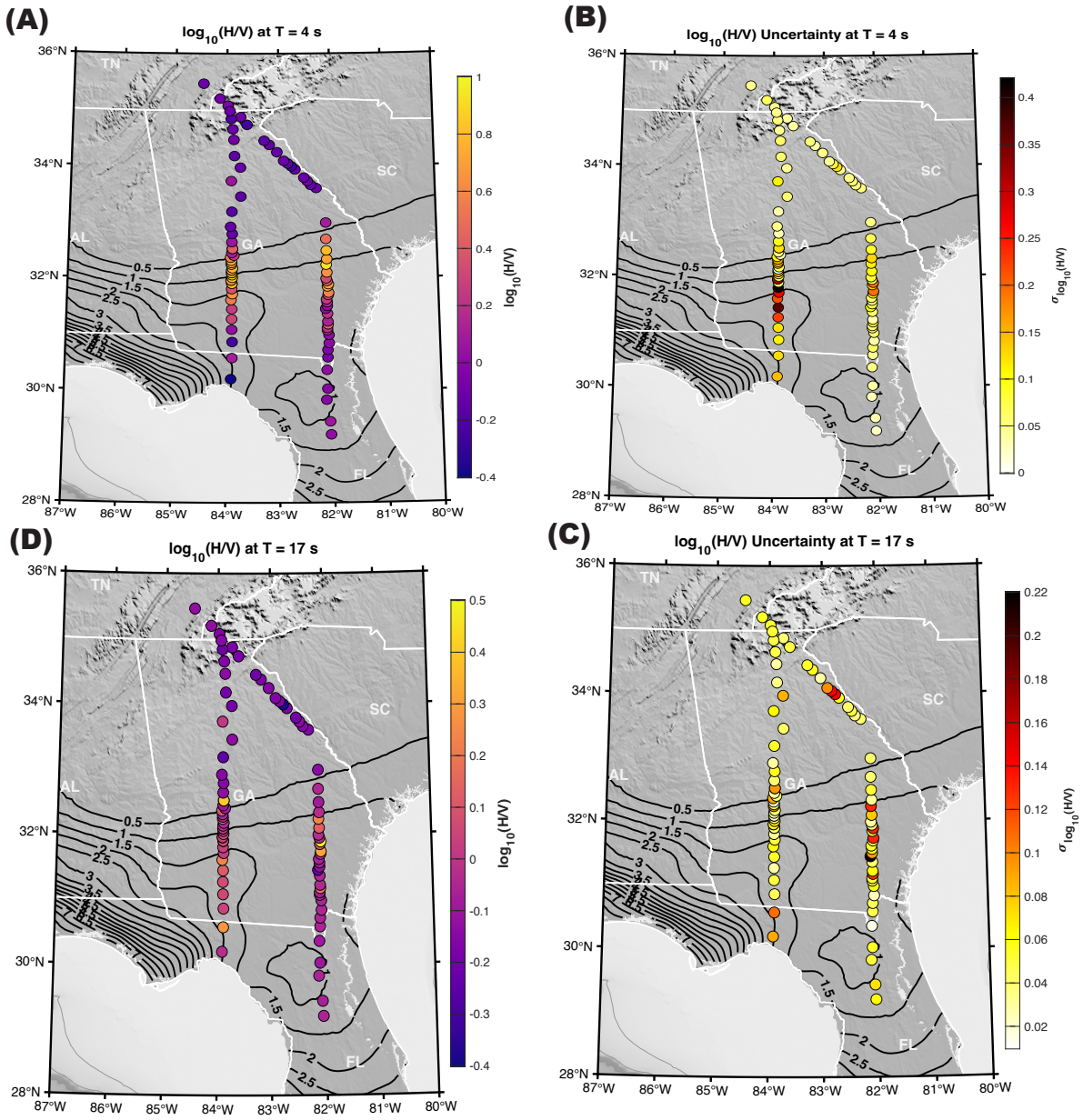

Figure 9: Fundamental-mode Rayleigh-wave ellipticity and associated uncertainty at 4 s (a, c) and 17 s (b, d) with sediment-thickness contours (km) from Boyd et al. (2024).

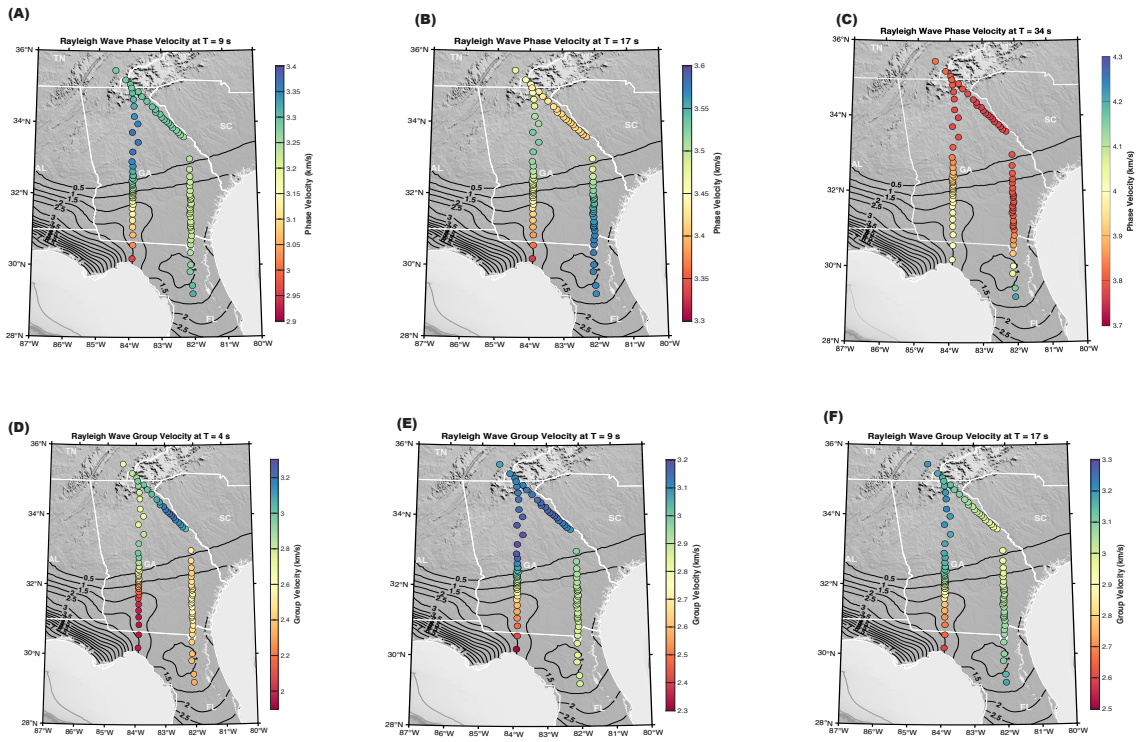

Figure 10: Rayleigh-wave phase velocity at T = 9, 17, and 34 s (a–c) and group velocity at T = 4, 9, and 17 s (d–f).

stiffer material that produces higher effective velocities than the unconsolidated column of the eastern ACP.

At 9 s (panel e), the range narrows to approximately 2.3–3.2 km/s as the group velocity sensitivity kernel deepens, with the SC transect showing a clear west-to-east decrease that still closely parallels the sediment-thickness contours. By 17 s (panel f), the range narrows further to approximately 2.5–3.3 km/s and the spatial pattern becomes more homogeneous, though some stations on the eastern transect near 30–31°N remain notably slower (∼2.6–2.7 km/s), suggesting that even at 17 s the deepest portions of the sedimentary column still influence the group velocity at these locations. The lowest group velocities we observe at short periods fall below the lower bounds reported in continental-scale tomographic studies (Bensen et al., 2007; Liang and Langston, 2008; Herrmann et al., 2021), which we attribute to the higher spatial resolution of the SESAME array that avoids smoothing the extreme velocity lows over the thickest portions of the sedimentary wedge.

Across both phase and group velocity measurements, the spatial gradient in velocity closely parallels the contours of sediment thickness, with the sharpest lateral transition occurring across the Fall Line where the sedimentary wedge thins to zero. This correspondence confirms that the observed dispersion is controlled primarily by the thickness and velocity structure of the Coastal Plain sediments, while stations on the Appalachians and Piedmont record velocities consistent with propagation through crystalline crust. We derived uncertainties in both phase and group velocity from the posterior model covariance of the tomographic inversion, extracted at the nearest resolved grid node to each station (Section 3.3.3; Appendix B). Per-station uncertainty maps at representative periods (supplementary material, Figure S6) show that most stations exhibit uncertainties well below 0.2% at short to intermediate periods, with elevated values only at the network extremities where ray coverage thins and at longer periods where fewer station pairs contribute reliable measurements. Together with the HVSR and ellipticity measurements, the dispersion data complete the suite of complementary constraints on shallow shear-wave velocity structure for joint inversion.

5 Discussion and Implications

We organize the discussion around three themes: the structural information revealed by the combined observables, the spectral convergence between HVSR and ellipticity that validates the theoretical framework, and a forward-modeling test demonstrating that a single velocity model simultaneously satisfies all four data types.

5.1 East–west structural divergence and controls on site resonance

The HVSR fundamental frequency f_0 varies by more than an order of magnitude across the SESAME array, and its spatial gradient closely parallels the sediment-thickness contours of Boyd et al. (2024), confirming that the sediment–basement impedance contrast acts as the primary control on the observed spectral ratios across the ACP. Our results agree with the thickness-dependent f_0 variations documented by Pratt (2018) and Pratt and Schleicher (2021) using both ambient-noise and earthquake spectral ratios, and with the inverse f_0–thickness relationship reported by Bodin and Horton (1999) and Bodin et al. (2001) in the Mississippi Embayment. The agreement between our ambient-noise HVSR and the earthquake-based estimates of Pratt (2018) provides empirical support for the argument we presented in Section 3.1.6: that the thick, heterogeneous ACP sedimentary column promotes sufficient wavefield randomization to satisfy the weaker form of the diffuse-field assumption, even in the presence of the strongly directional Atlantic Ocean noise source. This conclusion is further corroborated by the azimuthal HVSR analyses of Pratt (2018) and Rotimi et al. (2024), who examined the directional dependence of H/V spectral ratios across the ACP coastal plain sediments. Their results reveal no significant azimuthal variation in the HVSR, and the regional trend in the f_0 distribution is preserved across all back-azimuths, indicating that the ambient wavefield is sufficiently diffuse to produce stable site-response estimates independent of noise-source direction. This finding confirms that the f_0 pattern mapped in our study is governed by subsurface impedance structure rather than directional bias in the microseismic noise field.

Our most notable finding from the HVSR mapping is the east–west structural divergence, with the western portion of the array exhibiting anomalously high resonance frequencies (approaching ∼10 Hz) at stations where regional contours indicate a persistently thick sediment column of 1.5–2 km. This divergence implies that a single f_0–thickness calibration cannot transfer across the lithological transition from the Atlantic to the Gulf Coastal Plain. We interpret this pattern as evidence that HVSR in the western region responds to a shallower impedance contrast within the sedimentary column itself, where highly compacted or lithified units at depth act as an effective seismic basement above the true crystalline basement. Stations on the eastern and western transects occupy the same mapped Coastal Plain province, so the divergence records a difference in the internal structure and layering of the sedimentary column rather than in its mapped extent. This finding extends the regional understanding of ACP site response by demonstrating that the standard clastic-sediment model, which underpins most existing f_0–thickness regressions for the Atlantic margin (e.g., Pratt and Schleicher, 2021), breaks down toward the Gulf margin. Our dispersion results reinforce this structural distinction: short-period group velocity maps show slightly higher velocities along the western transect at equivalent latitudes, consistent with stiffer material in the western sedimentary column.

Although the present data cannot identify the responsible unit, the regional geology offers plausible candidates. Lithified Paleogene carbonates of the Floridan aquifer system lie at shallow depth beneath thin clastic overburden across much of southwestern Georgia (Miller, 1986; Renken, 1996), and lithified pre-Cretaceous strata and locally preserved basalt of the buried South Georgia Rift underlie the Coastal Plain section at greater depth (Chowns and Williams, 1983; Heffner et al., 2012); a sufficiently strong impedance contrast at the top of any of these units would register in HVSR as the effective seismic basement. Distinguishing among these candidates requires the depth-resolved velocity models from the joint inversion.

5.2 Spectral convergence of HVSR and Rayleigh-wave ellipticity

The spectral continuity between Rayleigh-wave ellipticity and microtremor HVSR provides direct observational support for the modal framework synthesized in Section 3.1.5. The waterfall comparison of Figure 11 demonstrates that the rising limbs of the ellipticity curves at the high-frequency edge of the measurement band (∼0.3 Hz) connect continuously to the onset of the HVSR resonance peaks, maintaining consistent

amplitude and shape in the overlap zone. This is the observational realization of the convergence predicted by Eq. (12): under diffuse-field conditions and fundamental-mode Rayleigh-wave dominance, HVSR reduces to Rayleigh-wave ellipticity, and the two quantities sample the same structural information from contiguous spectral windows. Several curves in Figure 11 display multiple peaks that migrate systematically along the transect; we interpret these as the expression of internal sedimentary layering whose character changes from south to north, registered coherently by both observables. Together the two observables span approximately three decades of frequency (0.04–25 Hz), providing continuous site-response characterization that motivates combining them with dispersion in a single inversion framework.

This result extends to a thick passive-margin sedimentary setting the relationship that Tanimoto et al. (2013) documented in the Los Angeles basin, but with an important quantitative difference. The amplitude offset between HVSR and ellipticity that we observed in the ACP consistently exceeds the ∼20% systematic difference reported by Tanimoto et al. (2013), with a median amplitude excess of ∼30% across the network within the 0.2–0.33 Hz overlap band of both observables (supplementary material, Figure S7). Much of this elevated median reflects the strong spatial gradient in sediment thickness across the array: stations in the northern ACP, where the sedimentary cover above the crystalline basement thins appreciably, record comparatively low H/V ratios, whereas stations in the southern ACP, where the sedimentary column thickens substantially, produce markedly higher H/V ratios that pull the network-wide distribution upward. We attribute the systematically larger HVSR-to-ellipticity discrepancy to the stronger Love-wave and body-wave contributions embedded in the full-wavefield HVSR, an effect amplified in the ACP by the efficient trapping and reverberation of horizontally polarized energy within the thick sedimentary column. This interpretation is consistent with the full modal expression of Eq. (10), which predicts that HVSR will exceed fundamental-mode ellipticity outside the band of strict Rayleigh-wave dominance. Crucially, this amplitude difference does not undermine the utility of jointly inverting the two observables; rather, it is precisely because HVSR and ellipticity sample different portions of the wavefield that they serve as complementary constraints. Our forward models honor this distinction by treating each data type according to its specific physical content, ensuring that the observed amplitude offset is accommodated rather than conflated.

5.3 Complementarity of observables and forward-modeling validation

All four observables are governed by the same underlying velocity structure, yet they sample it with different and complementary sensitivities (supplementary material, Figures S8–S10). HVSR constrains the depth to the principal impedance contrast through the resonance frequency, while ellipticity constrains the ratio of horizontal to vertical Rayleigh-wave eigenfunctions, which is sensitive to the sharpness and depth of impedance boundaries. Dispersion constrains the absolute velocities and their depth variation through the period-dependent phase and group speeds. Where these sensitivities overlap, the observables provide mutual validation: the spatial patterns in all three data sets track the sediment-thickness contours, and the period-dependent evolution of the ellipticity and dispersion maps is consistent with the frequency-dependent HVSR behavior. Where the sensitivities differ, the observables provide the complementary constraints necessary to reduce the non-uniqueness inherent in any single data type.

To illustrate this complementarity, we performed a preliminary joint forward-modeling test at a representative Coastal Plain station (W21), chosen to demonstrate consistency rather than for any anomalous behavior. We tested a five-layer velocity model consisting of three sedimentary layers (spanning 0 to approximately 1.5 km depth, with V_S stepping from approximately 0.5 km/s near the surface to approximately 2.0 km/s at the base) overlying two crustal layers extending to approximately 45 km depth (Figure 12). This single model reproduces the observed HVSR resonance peak position and amplitude near 1 s, captures the decay of ellipticity from H/V ≈ 3–4 at short periods toward values near unity at longer periods, fits the smooth phase velocity increase from ∼2.8 to ∼3.7 km/s, and tracks the overall group velocity trend including the low-velocity trough near 5–8 s that is characteristic of strong sedimentary influence. While HVSR and ellipticity occupy different but contiguous period bands (0.1–3 s and 3–25 s, respectively), the single velocity profile explains both, confirming the convergence predicted by Eq. (12). Equivalent tests at one station on the D transect and one on the E transect are shown in the supplementary material (Figures S11 and S12). This forward-modeling exercise is intended solely to demonstrate that the observables are mutually consistent; the formal joint inversion will employ a nonlinear hierarchical Bayesian framework with ensemble Markov chain Monte Carlo sampling, depth-dependent parameterization, full uncertainty quantification, and the flexibility

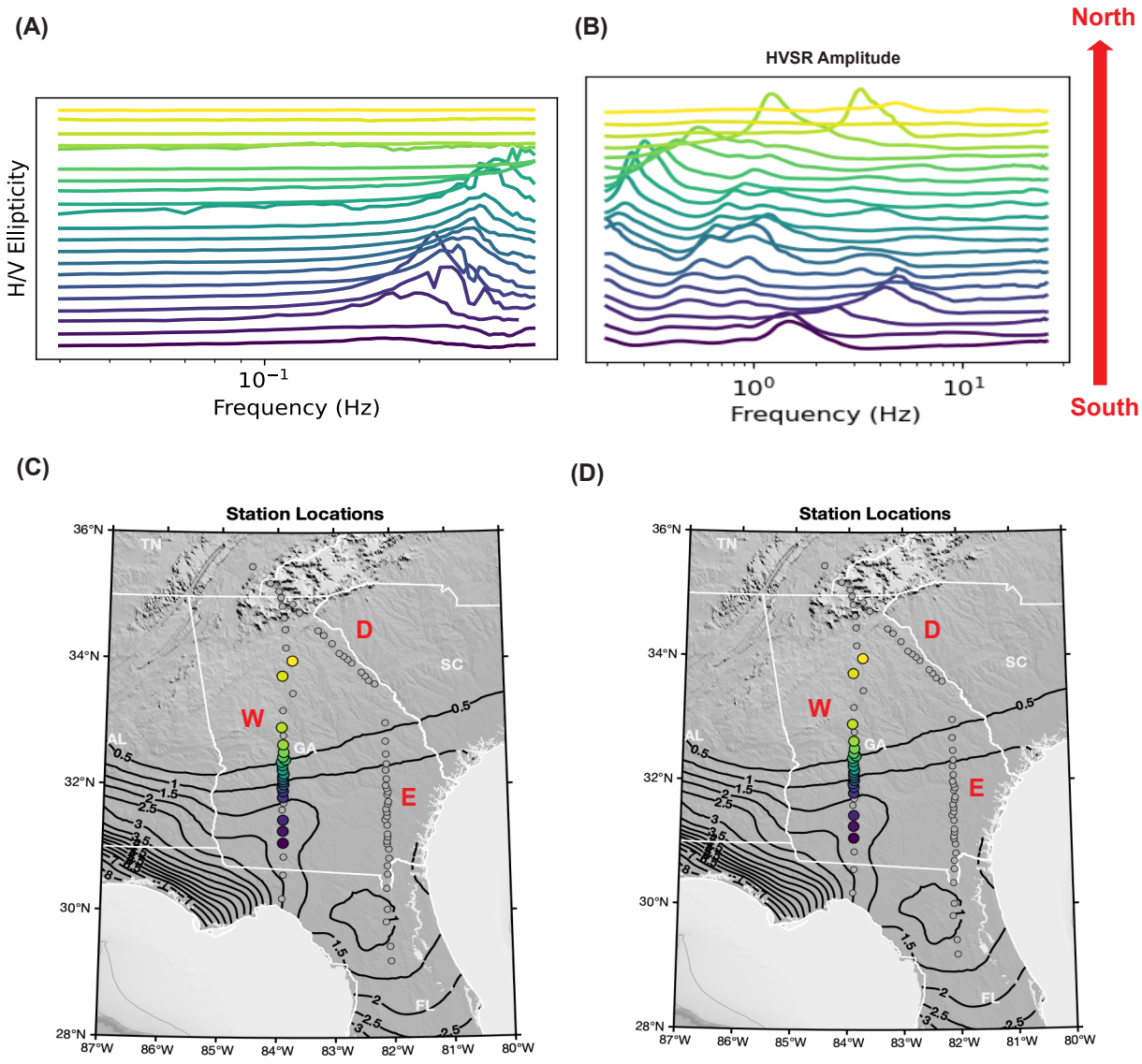

Figure 11: Waterfall comparison of Rayleigh-wave ellipticity (a) and microtremor HVSR (b) along the western transect, color-coded by latitude (south to north). Station locations shown in (c, d).

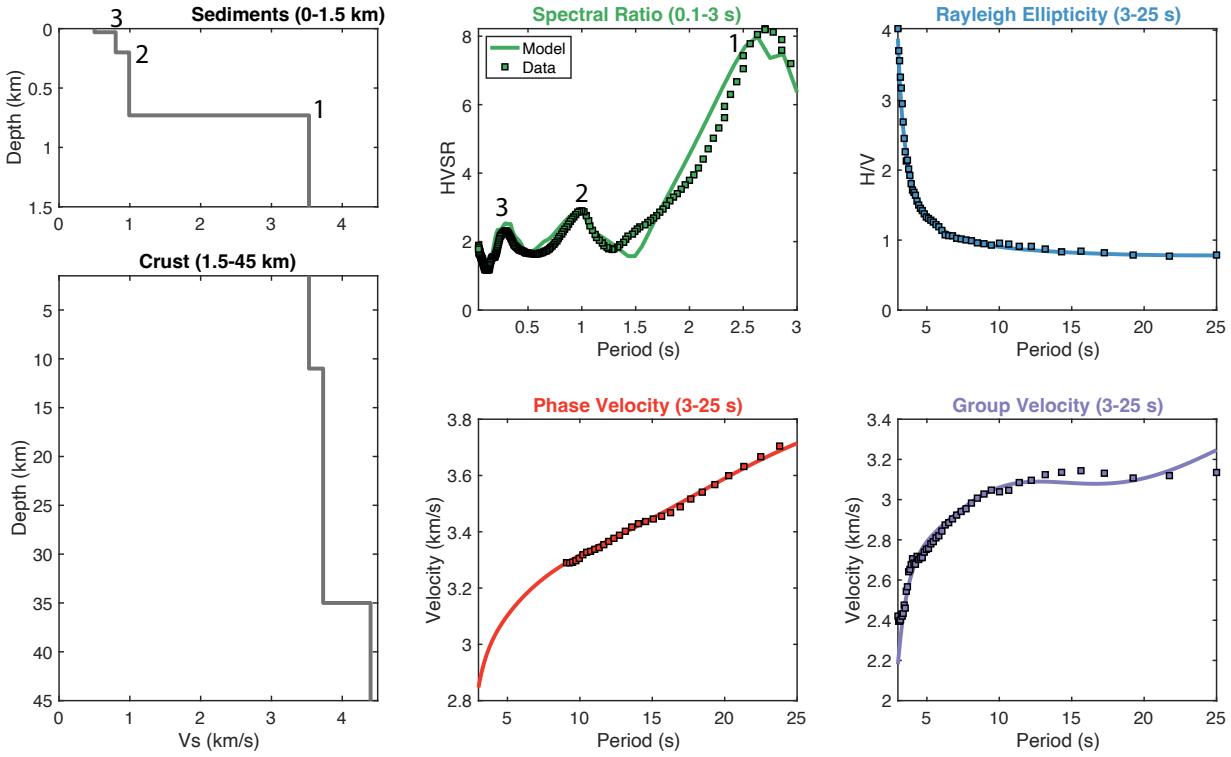

Figure 12: Preliminary joint forward modeling at a representative Coastal Plain station (W21): five-layer V_S model (left) and comparison of predicted (lines) versus observed (symbols) HVSR, ellipticity, phase velocity, and group velocity (right).

to accommodate laterally varying sediment properties across the array, because no single one-dimensional velocity model can satisfy the observables at every station.

5.4 Conceptual structural model and implications

Based on the combined data sets, we propose a conceptual structural model for the southeastern ACP and its transition to the Gulf Coastal Plain. In this model, the eastern ACP behaves as a classic passive-margin sedimentary wedge: a laterally continuous package of unconsolidated Cretaceous-to-Quaternary clastic sediments ($V_S \approx$ 200–1000 m/s) that thickens monotonically from zero at the Fall Line to more than 3 km in the southern embayment, overlying a sharp, first-order impedance contrast at the crystalline basement. The sediment column in this province is seismically slow, producing the strong HVSR peaks, elevated short-period ellipticities, and reduced group velocities that dominate our observations.

Toward the Gulf margin, the sedimentary column undergoes a change in mechanical properties at depth, the presence of highly compacted or lithified units (e.g., Chowns and Williams, 1983; Miller, 1986) introduces a secondary impedance boundary within the sediment column that, for the frequencies sampled by HVSR, acts as the effective acoustic basement. This partitions the column into an upper unconsolidated section and a lower, stiffer section. We hypothesize that the joint inversion will recover this internal structure as a pronounced velocity step within the sedimentary column at western stations, absent at eastern stations, and that the depth and velocity contrast at this boundary may correlate with the mapped extent of the Suwannee terrane and the Paleozoic suture that separates Laurentian from peri-Gondwanan basement beneath the Coastal Plain (Hatcher et al., 2007; Hibbard et al., 2010).

If confirmed, this would establish a direct link between the shallow seismic response measured at the surface and the deep tectonic architecture inherited from the Appalachian orogeny, with practical implications

for seismic hazard assessment in the rapidly urbanizing southeastern United States. Current site amplification models assume laterally uniform sediment properties across the Coastal Plain (Boyd et al., 2024; Gann-Phillips et al., 2024), and result from our observables suggest that this assumption is not fully justified across the Atlantic-to-Gulf transition. The consistency of our observations across the study area spanning more than six degrees of latitude and encompassing sediment thicknesses from zero at the Fall Line to more than 10 km along the southern coast demonstrates that the passive seismic approach we employed here characterizes the sedimentary structure at a resolution exceeding what continental-scale tomographic models achieve alone.

6 Conclusions

This study presents a passive seismic characterization of the Atlantic Coastal Plain and southern Appalachians using three complementary observables (microtremor HVSR, fundamental-mode Rayleigh-wave ellipticity, and surface-wave phase and group velocity dispersion) extracted from continuous broadband recordings of the 88-station SESAME array. To our knowledge, this is the first regional study in the southeastern United States to combine these passive seismic techniques within a unified theoretical framework under the diffuse-field assumption.

We showed that strict equipartition is not required for HVSR to carry structural information in the ACP; sufficient azimuthal randomization of the surface-wave field, promoted by trapping, scattering, and reverberations within the sedimentary wedge, satisfies the weaker diffuse-field condition. Following the modal decomposition of Sánchez-Sesma et al. (2011) and García-Jerez et al. (2016), we showed that HVSR reduces to fundamental-mode Rayleigh-wave ellipticity under conditions met near the principal site resonance, and we confirmed this prediction observationally through waterfall comparisons demonstrating spectral continuity between the two observables near 0.3 Hz. Together, HVSR and ellipticity provide continuous site-response characterization from approximately 0.04 to 25 Hz.

The combined observables reveal two principal findings. First, HVSR peak frequencies span more than an order of magnitude across the array and track sediment-thickness contours on the eastern transects, confirming that the sediment–basement impedance contrast is the primary control on site resonance. On the western array, however, anomalously high resonance frequencies despite thick sediments reveal an east–west structural divergence that we attribute to the presence of stiffer material within the sedimentary column acting as an effective seismic basement. This finding demonstrates that the standard clastic-sediment model underpinning existing f_0–thickness regressions for the Atlantic margin does not hold toward the Gulf margin. Second, new high-resolution phase and group velocity maps resolve the sharp velocity gradient across the Fall Line at finer spatial detail than continental-scale tomographic models, with short-period group velocities revealing a north–south contrast exceeding 1 km/s. Preliminary forward modeling confirmed that a five-layer parameterization simultaneously reproduces all four observables at a representative Coastal Plain station, validating the theoretical complementarity.

These regionalized dispersion curves and station-level HVSR and ellipticity measurements provide the observational foundation for Part 2, where a nonlinear hierarchical Bayesian inversion will recover the shear-wave velocity structure from surface to upper mantle with full uncertainty quantification, addressing the need for well-constrained velocity models that span both the Appalachian crystalline terrane and the adjacent Coastal Plain for improved seismic hazard assessment in the southeastern United States.

A Details of the dispersion measurements

Each narrow-band filter used in the frequency–time analysis of Section 3.3.2 takes the form:

$$G(\omega, \omega_0) = \exp\left[-\left(\frac{\omega - \omega_0}{\sigma_\omega}\right)^2\right], \tag{A1}$$

where ω_0 is the center angular frequency and σ_ω is the $1/e$ half-width, held constant across all center frequencies.

We derived the uncertainties of the Bessel-function phase velocity fits (Section 3.3.1) from the model covariance matrix of the nonlinear least-squares problem. We estimated the data variance as:

$$\sigma_d^2 = \frac{\mathbf{e}^T\mathbf{e}}{N_d - N_m}, \tag{A2}$$

where $\mathbf{e}$ is the residual vector, N_d is the number of data points, and N_m is the number of model parameters. The model covariance matrix is:

$$\mathbf{C}_m = \left(\mathbf{J}^T\mathbf{J}\right)^{-1}\sigma_d^2, \tag{A3}$$

where $\mathbf{J}$ is the Jacobian at the converged solution. We propagated the travel-time uncertainties (square root of the diagonal of $\mathbf{C}_m$) to phase velocity through:

$$\sigma_c = r\,t^{-2}\,\sigma_t. \tag{A4}$$

B Tomographic inversion formulation

We formulated the tomographic problem of Section 3.3.3 as an augmented damped least-squares system:

$$\begin{bmatrix} \mathbf{WG} \\ \lambda_s\mathbf{F} \\ \lambda_d\mathbf{D}_{\mathrm{azi}} \\ \lambda_{s,\mathrm{azi}}\mathbf{F}_{\mathrm{azi}} \end{bmatrix} \mathbf{m} = \begin{bmatrix} \mathbf{Wd} \\ \mathbf{0} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \tag{B1}$$

where $\mathbf{G}$ is the ray-theory data kernel relating slowness to traveltime, constructed from great-circle ray paths between station pairs with each ray discretized into 100 waypoints and the mean azimuth along the path used for the anisotropic terms; $\mathbf{W}$ is the diagonal data weighting matrix; $\mathbf{d}$ is the observed traveltime vector; and $\mathbf{m}$ contains the isotropic slowness values together with the two azimuthal anisotropy coefficient fields (A_{c2}, A_{s2}). We applied three classes of regularization: second-derivative smoothing $\mathbf{F}$ on the isotropic field penalizes rough lateral velocity variations, norm damping $\mathbf{D}_{\mathrm{azi}}$ on the anisotropic parameters suppresses spurious anisotropy where ray coverage is sparse, and separate second-derivative smoothing $\mathbf{F}_{\mathrm{azi}}$ on the anisotropic fields enforces spatial coherence. We set the base regularization weights to $\lambda_{s,0} = 1$ for isotropic smoothing and $\lambda_{d,0} = \lambda_{s,\mathrm{azi},0} = 10^3$ for the anisotropic terms, and at each period we rescaled these weights by the ratio of the data kernel norm to the corresponding regularization kernel norm (e.g., $\lambda_s = \lambda_{s,0}\,\|\mathbf{WG}\|_1/\|\mathbf{F}\|_1$) so that the effective regularization strength adapts automatically to the period-dependent data density.

We weighted the data by the inverse of the fractional traveltime error, $W_{ii} = \left(\sigma_t^{(i)}/t^{(i)}\right)^{-1}$, where σ_t is the traveltime uncertainty propagated from the phase or group velocity standard deviation and t is the observed traveltime, and we capped the weights at the 25th percentile of the fractional error distribution to prevent a small number of unusually precise measurements from dominating the solution. We identified gross outliers by fitting a linear trend to the traveltime–distance relationship at each period and downweighted to zero any measurement whose residual exceeded 2 s. After the initial inversion, a single iteration of residual-based rejection removed measurements with residuals exceeding two standard deviations of the residual distribution, after which we recalculated the regularization weights and re-inverted the system. We masked as unresolved any grid node sampled by fewer rays than a minimum density threshold, set at 0.25 times the grid spacing in kilometers, and excluded such nodes from subsequent analysis.

We estimated model uncertainties at each grid node from the diagonal of the posterior model covariance matrix. Scaling the data rows of the augmented system by the inverse of the observed traveltimes converts from slowness to velocity units and yields the velocity uncertainty directly:

$$\sigma_{\mathbf{m}} = \sqrt{\mathrm{diag}\left[\left(\mathbf{A}_t^T\mathbf{A}_t\right)^{-1}\right]}, \tag{B2}$$

where $\mathbf{A}_t$ denotes the rescaled augmented system matrix. We characterized model resolution by the diagonal of the resolution matrix $\mathbf{R} = \left(\mathbf{A}^T\mathbf{A}\right)^{-1}\mathbf{G}^T\mathbf{W}^2\mathbf{G}$ and computed the normalized χ^2 misfit at each period to assess the overall quality of the data fit. For the checkerboard tests summarized in Section 3.3.3, we computed synthetic phase-velocity traveltimes from the actual station-pair geometry and inverted them with the same regularization parameters and quality control thresholds applied to the real data, while for group velocity we extracted the ray geometry directly from the tomographic output.

Data Availability Statement

All broadband waveform data were accessed through the IRIS Data Management Center under network code Z9 using the open-source Python package ObsPy (`https://docs.obspy.org/`). The script used for retrieving daily waveform data can be found at `https://github.com/jbrussell/fetch_NOISE`. Codes used for calculating ambient-noise cross-correlations and extracting phase velocities can be found at `https://github.com/jbrussell/MATnoise`. The HVSR quality control script is accessible at `https://github.com/RotimiBabatundeIsaac/SESAME_DC`. All figures were created using Generic Mapping Tools software (Wessel et al., 2013) and MATLAB plotting tools. HVSR under the diffuse field assumption was computed using the software HVSRpy by J. P. Vantassel (`https://github.com/jpvantassel/hvsrpy`).

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